



Cyanobacteria as biochemical treasure troves: unveiling bioactive metabolites for pharmaceutical and global sustainability

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Abstract

Cyanobacteria, among the most ancient and versatile microorganisms on Earth, thrive across different aquatic ecosystems, ranging from freshwater to marine environments. In addition to being important producers of structurally diverse secondary metabolites, these photosynthetic prokaryotes are crucial to the global cycling of carbon and nitrogen. With their wide range of biological activities, such as antibacterial, antiviral, anticancer, and anti-inflammatory properties, cyanobacterial metabolites are attractive options for use in biotechnology and medicine. In addition, some cyanometabolites have toxic properties that affect aquatic ecosystems and present problems for human health and water quality. This dual significance for ecology and biomedicine emphasizes how crucial it is to do systematic research on cyanobacterial metabolites. The chemical diversity, biological roles, and biosynthetic origins of cyanobacterial secondary metabolites are summarized in this review, with a focus on their translational potential and ecological responsibilities. Innovative approaches for accelerating metabolite discovery and characterization are also covered, including recent developments in metabolite profiling, genome mining, and artificial intelligence (AI)-assisted biosynthetic gene cluster prediction. The present review integrates ecological, biochemical, and computational perspectives emphasizing the significance of cyanobacteria as sources of bioactive chemicals for environmental sustainability, pharmaceutical research, and biotechnology.

Keywords Artificial intelligence · Biological properties · Cyanobacteria · Cyanotoxins · Metabolites · Pharmaceuticals

Introduction

Cyanobacteria, sometimes known as blue-green algae, are among the oldest and most ecologically important types of photosynthetic prokaryotes, having evolved over 3.5 billion years (Kharwar and Mishra 2024). Modern biogeochemical

cycles and the rise of aerobic life forms were made possible by these microbes' essential contribution to the oxygenation of Earth's atmosphere (Mazard et al. 2016; Singh et al. 2016; Zhang et al. 2023). Cyanobacteria are still essential to the worldwide cycling of carbon and nitrogen today, and they are found in every possible habitat such as freshwater, marine, and terrestrial environments.

Cyanobacteria are oxygenic phototrophs, and their main pigment for photosynthetic activity is chlorophyll *a* (Kharwar and Mishra 2020). They also have accessory pigments including phycocyanin, phycoerythrin, and carotenoids, which work together to improve light-harvesting efficiency and photoprotection (Solanki 2020; Kuczyńska et al. 2024; Minić 2021). From filamentous and heterocyst-forming taxa to unicellular and colonial forms, cyanobacteria have evolved into a vast variety of morphotypes during the course of evolution. They have been able to colonize an astounding variety of ecological niches because of their structural and physiological diversity, including settings with high UV radiation, desiccation, poor light availability, nutrient constraint, and extreme temperatures (Rippka et al. 1974; Saw

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et al. 2013; de la Rosa et al. 2024). Significantly, cyanobacteria exhibit adaptability by maintaining photosynthetic activity in environments that severely restrict the majority of light-dependent species (Foster et al. 2011; Maberly And Gontero 2022; Steunou et al. 2006; Velichko et al. 2021).

In addition to their ecological significance, cyanobacteria are becoming more widely known for their extraordinary metabolic adaptability (Dittmann et al. 2015; Singh et al. 2017; Baunach et al. 2024; Jeong et al. 2020). They promote the synthesis of a wide variety of primary and secondary metabolites because they are effective photosynthetic organisms with better solar-to-biomass conversion efficiencies than many terrestrial plants (Ducat et al. 2011; Kanno et al. 2017; Olowoyeye And Kanwar 2023). Secondary metabolites are important for ecological interactions such as stress tolerance, metal chelation, signaling, symbiosis, and defense against grazers and competition, whereas primary metabolites are necessary for growth and cellular maintenance (Singh et al. 2017; Burja et al. 2001; Manikandan et al. 2022). These compounds are commonly synthesized through specialized biosynthetic gene clusters and encompass chemically diverse classes such as peptides, polyketides, alkaloids, terpenoids, fatty acids, polysaccharides, and photoprotective pigments (Srivastava et al. 2022; Shih et al. 2013; Kehr and Dittmann 2015; Ahmad et al. 2022).

Cyanobacterial secondary metabolites have a wide range of biological activities, such as antibacterial, antiviral, anticancer, anti-inflammatory, and enzyme-inhibitory qualities, which has drawn ongoing attention from the biotechnology and pharmaceutical industries (Singh et al. 2005, 2016; Ahmad and Singh 2024; De Obeso Fernandez Del Valle And Scheckhuber 2021; Tiwari And Tiwari 2020). At the same time, some cyanobacterial metabolites are strong poisons that can cause toxic algal blooms and endanger aquatic ecosystems, drinking water quality, and individual health (Gupta et al. 2023; Pierobon et al. 2018; Li et al. 2014; Singh et al. 2023a, b, c; Campos And Vasconcelos 2010). Comprehensive research into the diversity, function, and control of cyanobacterial metabolites is crucial because of their dual nature, which has the potential to be both medicinal and ecologically hazardous (Yasin et al. 2021).

Many reviews of cyanobacterial metabolites have been conducted in recent decades, frequently concentrating on particular chemical classes (such toxins or peptides), specific biological activity, or well-researched taxa (Alves et al. 2020; Cepoi 2024; Leao et al. 2012; Köcher et al. 2020; Li et al. 2020; Nawaz et al. 2023; Okeke et al. 2021; Osborne et al. 2007; Osborne And Shaw 2008; Welker and Döhren 2006). There are still significant gaps in the knowledge connecting biosynthetic gene clusters to ecological function in natural environments, despite the thorough characterisation

of cyanobacterial metabolites. In order to better understand metabolite diversity in understudied taxa and harsh settings, the majority of studies focus on laboratory-cultured strains and well-studied species. In order to translate the chemical variety of cyanobacteria into sustainable pharmacological and biotechnological uses, future research must integrate ecological and evolutionary frameworks with metabolomics, genome mining, and AI-assisted prediction. The current literature is still quite dispersed, with little synthesis across metabolite classes, ecological roles, biosynthetic pathways, and new technological developments, despite the fact that these investigations have yielded insightful information. The analytical and conceptual framework available for cyanobacterial natural product research has also been significantly expanded by recent developments in metabolite profiling, genome mining, and artificial intelligence (AI)-assisted biosynthesis pathway prediction, necessitating an updated and comprehensive reassessment.

In light of this, the current study attempts to give a comprehensive and multidisciplinary overview of cyanobacterial bioactive metabolites. This review provides an integrative approach, relating functional metabolite diversity, biosynthetic gene cluster architecture, and ecological adaptation. The review specifically seeks to: (i) deliver an overview of the main classes of secondary metabolites produced by cyanobacteria and their biosynthesis origins; (ii) discuss the biological activities and ecological roles of these metabolites in a unified framework; and (iii) highlight new analytical, computational, and artificial intelligence (AI)-assisted methods for metabolite discovery and characterization. In order to promote future research and translational efforts in pharmaceutical development, biotechnology, and environmental sustainability, this review provides a thorough and forward-looking synthesis.

Secondary metabolites produced by cyanobacteria

Non-ribosomal peptides synthase (NRPS), polyketides synthase (PKS), and hybrid PKS-NRPS pathway

NRPS and PKS are key enzymes involved in the biosynthesis of secondary metabolites across diverse organisms (Shukla And Sharma 2024; Soohoo et al. 2024; Gademann And Portmann 2008; Li et al. 2020; Weber et al. 2015). These enzymes are characterized by at least three modular domains that facilitate chain elongation and molecular transformations (Ansari et al. 2004). Initially, the catalytic domain binds and activates the precursor substrate, which is then transferred to the carrier protein domain. The carrier

protein domain subsequently loads the growing PK or NRP chain, incorporating the activated precursors. Finally, another catalytic domain mediates bond formation between the growing chain and newly loaded building blocks. A key distinction between PKS and NRPS lies in their use of precursor molecules: PKS utilizes malonyl-CoA or methyl malonyl-CoA, whereas NRPS incorporates proteinogenic and non-proteinogenic amino acids as monomers. Multiple genes make to normal PKS. Each module in the genes is typically in charge of adding and processing a single component in the developing chain. Each domain that makes up the modules catalyses one or a few chemical changes on the subunit. Protein sequences frequently contain these well-characterized domains that are simple to identify. Modules inside NRPS are in charge of incorporating particular amino acids into peptide chains (Yang et al. 2022). These modules include both proteinogenic and nonproteinogenic amino acids and are composed of an adenylation domain, peptidyl carrier domain, and condensation domain. The complex chemical structures created by further changes including N-methylation, epimerization, and cyclization of the amino acid backbone can also occur in other domains.

Interestingly, hybrid PKS-NRPS systems have also been identified, as capable of synthesizing complex compounds. Notably, toxins such as lyngbyatoxin, apratoxin, and aplysiatoxin are synthesized through PKS, NRPS, or hybrid PKS-NRPS pathways. NRPS systems also produce lipopeptides, which are characterized by a lipid-linked peptide structure formed through the combination of a lipid tail and amino acids. For instance, *Moorea producens* synthesizes antillatoxin and carmabin, while *Moorea bouillonii* produces lyngbyabellin. These lipopeptides display diverse bioactivities, with carmabin exhibiting antimalarial properties, and antillatoxin and lyngbyabellin demonstrating potent neurotoxic and cytotoxic effects. Moreover, siderophores are small, high-affinity iron-chelating molecules synthesized by NRPS are crucial for bacterial survival in iron-limited environments. Since iron predominantly exists in an insoluble form, certain bacteria have evolved to secrete siderophores to enhance iron absorption, underscoring their critical role in microbial metabolism and adaptability.

Terpenoids

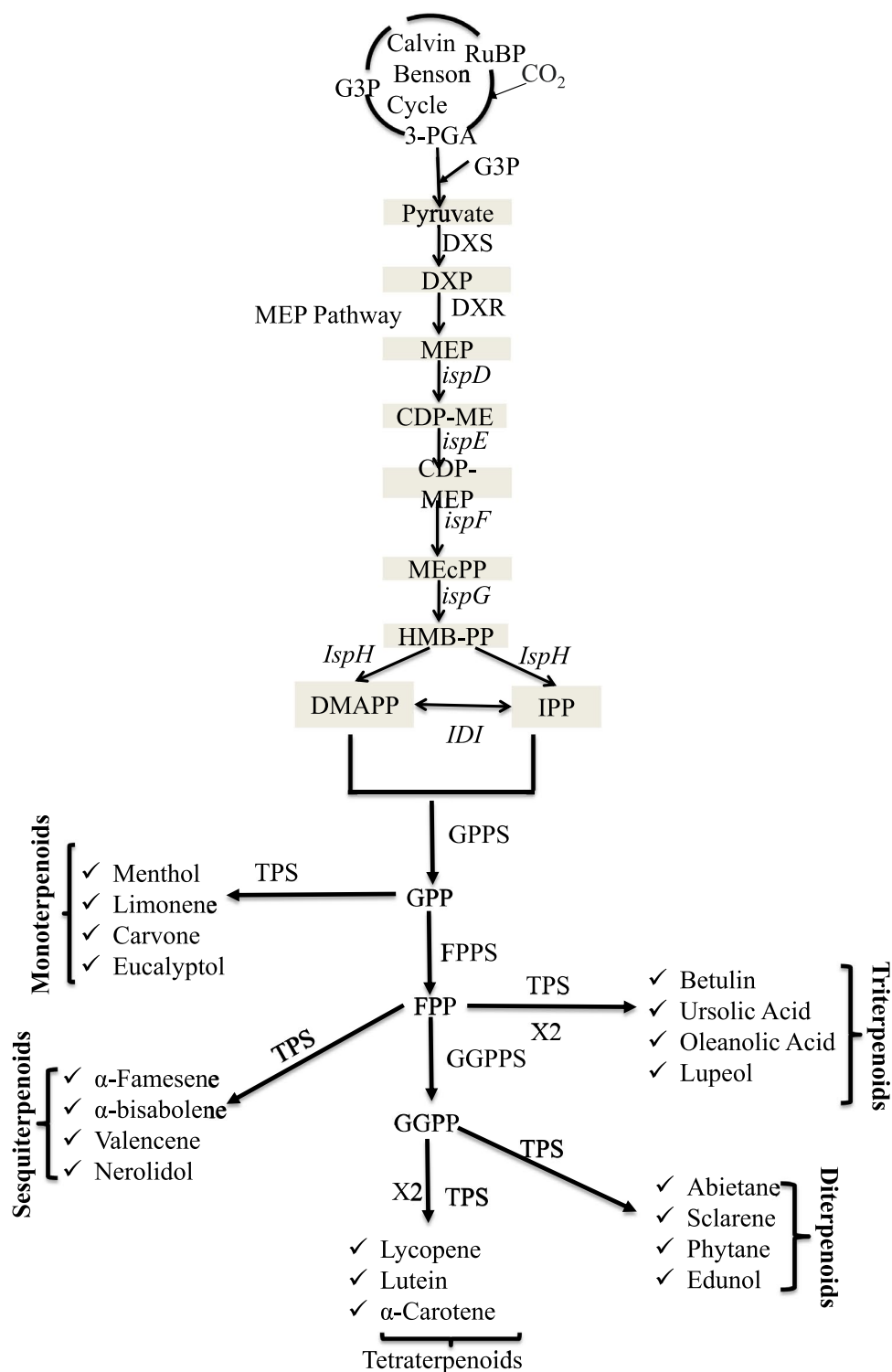
Terpenoids represent a diverse class of organic compounds that are abundantly distributed in cyanobacteria and algae (Hoshino 2024; Husain et al. 2024; Pattanaik And Lindberg 2015; Vincent et al. 2023; Avalos et al. 2022; Gupta And Phulara 2021). They can be categorized into seven groups: hemiterpenoids, monoterpenes, sesquiterpenes, diterpenes, triterpenes, tetraterpenes, and polyterpenes based on their structural foundation of five-carbon isoprene units

(Keeling And Bohlmann 2012; Singh And Sharma 2015). Terpenes encompass a broad spectrum of natural products (Gershenzon And Dudareva 2007) and are synthesized by plants, fungi, and bacteria through two primary pathways: the mevalonate (MVA) pathway and the methylerythritol phosphate (MEP) pathway (Fig. 1) (Gao et al. 2016). These biosynthetic pathways utilize substrates such as acetyl-CoA or glyceraldehyde 3-phosphate and pyruvate (Pattanaik And Lindberg 2015). While the MVA and MEP pathways often operate synergistically across various organisms, cyanobacteria predominantly rely on the MEP pathway, leveraging substrates generated during photosynthesis (Lee et al. 2017). This MEP pathway produces isomeric five-carbon compounds, specifically isopentyl pyrophosphate and dimethylallyl pyrophosphate, which are subsequently condensed into geranyl pyrophosphate (GPP), a crucial precursor in terpene biosynthesis (Rautela and Kumar 2022). The versatile structure of GPP enables the production of numerous terpenes, highlighting the significance of cyanobacteria as a source of diverse terpenoid compounds (Li et al. 2014; Noushahi et al. 2022; Randhir et al. 2020; Ruzicka 1953; Żymańczyk-Duda et al. 2022).

Terpenes play a crucial role in various cellular processes essential for survival (Rogowska And Szakiel 2020). They are involved in energy metabolism through ubiquinone, which acts as a key energy carrier during cellular respiration. Additionally, terpenes contribute to photosynthesis *via* compounds such as chlorophyll, carotenoids, and plastoquinone, while hopanoids are vital for maintaining cell membrane biosynthesis and stability (Belin et al. 2018; Nandagopal et al. 2021). Among the diverse array of terpene compounds, carotenoids derived from cyanobacteria are of significant commercial interest due to their different applications. Notably, certain cyanobacterial species exhibit allelopathic properties, which inhibit the growth of neighboring cyanobacterial species, primarily attributed to the production of substantial amounts of geranyl acetone (Fischer 1991; Li et al. 2023).

Cyanobacterial terpenoids are involved in redox metabolism, membrane integrity, and photosynthesis, among other vital cellular functions. Carotenoids reduce photooxidative damage to photosynthetic machinery by dissipating excess excitation energy and quenching reactive oxygen species (Sun et al. 2022; Zulfiqar et al. 2021). Hopanoids help make membranes stable and stiff, especially when they are stressed by the environment (Kulkarni et al. 2015). Additionally, some terpenoid derivatives have allelopathic properties that prevent the growth of rival microbes (Kumar et al. 2024; Khalifa et al. 2021).

Fig. 1 Diagrammatic illustration of terpenoid production in cyanobacteria. The isoprenoid precursors isopentenyl pyrophosphate (IPP) and dimethylallyl pyrophosphate (DMAPP) are produced by the methylerythritol phosphate (MEP) route. The larger prenyl diphosphates (GPP, FPP, and GGPP) that result from the conversion of these intermediates are precursors to carotenoids and other isoprenoid-derived metabolites that are involved in membrane function, photosynthesis, and photoprotection. *RuBP* ribulose-1,5-bisphosphate, *3-PGA* 3-phosphoglyceric acid, *Ac-CoA* acetyl-CoA, *AcAc* CoA acetoacetyl-CoA, *HMG-CoA* 3-hydroxy-3-methylglutaryl-CoA, *MVAmevalonate*, mevalonate-5-phosphate, *MVAPP* mevalonate-5-pyrophosphate, *G3P* glyceraldehyde 3-phosphate, *DXP* 1-deoxy-d-xylulose 5-phosphate, *MEP* methylerythritol-4-phosphate, *CDP-ME* 4-(cytidine 5'-diphospho)-2-C-methyl-d-erythritol, *CDPMEP* 2-phospho-4-(cytidine 5'-diphospho)-2-C-methyl-d-erythritol, *MEcPP* 2-C-methyl-d-erythritol 2,4-cyclo-diphosphate, *HMB-PP* 4-hydroxy-3-methylbut-2-enyldiphosphate, *AtoB* acetoacetyl-CoA thiolase, *HMGS* HMG-CoA synthase, *HMGR* HMG-CoA reductase, *MK* mevalonate kinase, *PMK* MVAP kinase, *MVAPP* decarboxylase, *DXS* DXP synthase, *DXR* DXP reductoisomerase, *IspD*: CDP-ME cytidyltransferase, *IspE* CDP-ME kinase, *IspF* MEC synthase, *IspG* HMBPP synthase, *IspH* HMBPP reductase, *DMAPP* dimethylallyl diphosphate, *IPP* isopentenyl diphosphate, *GPP* geranyl diphosphate, *FPP* farnesyl diphosphate, *GGPP* geranylgeranyl diphosphate, *IDI* isopentenyl diphosphate isomerase, *GPPS* GPP synthase, *FPPS* FPP synthase, *GGPPS* GGPP synthase, *TPS* Terpene synthase



Carotenoids

The tetraterpenoid pigments known as carotenoids are synthesised from units derived from isoprene and have a C40 polyene backbone with several conjugated double bonds that give them their antioxidant and light-absorbing

qualities (Nisar et al. 2015; Armstrong et al. 1990; Liang et al. 2006; Manochkumar et al. 2022; Pereira et al. 2021). These accessory pigments play a vital role in photosynthesis, absorbing light primarily in the 400 to 500 nm range, with their spectral properties influenced by the number and arrangement of double bonds (Bramley

And Mackenzie 1988; Solomons And Bulux 1993; Yuan et al. 2015). Two prominent carotenoids, beta-carotene, and echinenone, are particularly important within this group (Pagels et al. 2021). Other significant carotenoid compounds found in cyanobacteria and algae include astaxanthin, cryptoxanthin, zeaxanthin, canthaxanthin, and 30-hydroxyechinenone which can be synthesized by process depicted in Fig. 2 (Mochimaru et al. 2005; Shah et al. 2016; Singh et al. 2017). These carotenoids serve multiple functions, acting as colorants, precursors to visual pigments, and enhancing the antioxidant capacity in plants and algae as well as in cyanobacteria (Kharwar et al. 2022; Čmiková et al. 2024; Maschmeyer et al. 2020). By scavenging reactive oxygen species (ROS) and dissipating excess energy, these pigments protect the photosynthetic apparatus (Mohamed Abdoul-Latif et al. 2024). One of the most well researched carotenoids, β -carotene i.e., most biologically active form, leads to vitamin A derivatives like retinal and retinoic acid, which are critical for immune system function, cellular differentiation, and vision in higher plants (Olson 1993; Seino et al. 2008). The ecological significance of cyanobacterial carotenoids and their prospective uses in biotechnological, pharmacological, and nutraceutical sectors are highlighted by their functional variety (Varshney and Singh 2013).

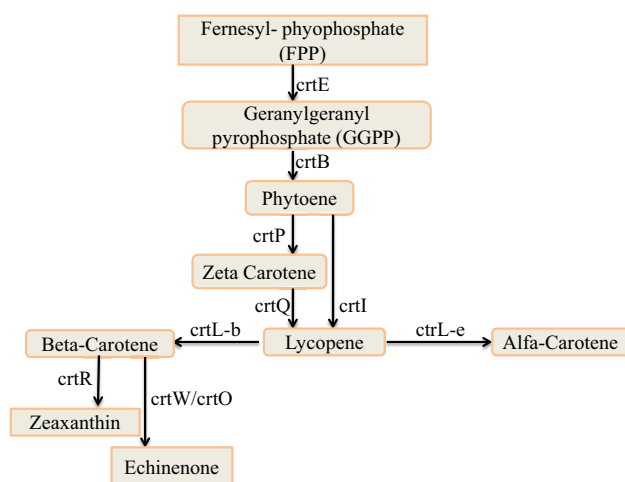


Fig. 2 Diagrammatic representation of steps involved in the production of carotene in cyanobacteria. Geranylgeranyl pyrophosphate (GGPP), which is produced *via* the MEP route, is the starting point for the biosynthesis of carotene. GGPP condensation results in phytoene, that subsequently goes through a series of desaturations to yield lycopene and cyclization to yield β -carotene and other carotenoids that help in photosynthesis and photoprotection. *crtE* geranylgeranyl pyrophosphate synthase, *crtB* phytoene synthase, *crtP* phytoene desaturase, *crtQ* zeta-carotene desaturase, *crtL* lycopene beta cyclase, lycopene epsilon cyclase, *cruA* lycopene cyclase, *crtO/crtW* beta-carotene ketolase, *crtR* beta-carotene hydroxylase

Sterols

Sterols play a crucial role in regulating membrane fluidity, a fundamental property that affects cellular function (Piironen et al. 2000; Silvestro et al. 2013; Volkman 2003). These compounds are synthesized as byproducts of isoprenoid biosynthesis. It has been reported that only few cyanobacterial species have the genes necessary for sterol biosynthesis and can produce sterols. This suggests that sterol production is limited to particular lineages rather than being a characteristic shared by all cyanobacteria (Volkman 2003). Cyanobacteria are capable of producing both saturated and unsaturated sterols, as noted by Kohlhasse and Pohl (1988). The unsaturated sterols identified in cyanobacteria include cholesterol, chondrillasterol, stigmaterol, sitosterol, brassicasterol, campesterol, 22-dehydrocholesterol, 24-ethyl-cholest-7-enol, 24-methyl-cholest-7-enol, 24-ethylcholesta-2,5-dienol, and 24-ethylcholesta-5,7,22-trienol (Elsheikh And Eltanahy 2024; Forin et al. 1972; Levin And Bloch 1964; Nadal 1971; Paoletti et al. 1976; Perry et al. 1978; Reitz And Hamilton 1968; Seckbach And Ikan 1972; Singh et al. 2017; de Souza and Nes 1968; Teshima And Kanazawa 1972). In contrast, several saturated sterols, including 4 α -methylsterols, 5 α -cholestan-3 β -ol, 24-methyl-5 α -cholestan-3 β -ol, and 24-ethyl-5 α -cholestan-3 β -ol, have been reported in cyanobacterial species such as *Anabaena cylindrical*, *A. viguirei*, *A. solitaria*, *Nostoc carneum*, *Nodularia harveyana*, and *Microcystis aeruginosa* (Nishimura And Koyama 1977; Patterson 1971; Singh et al. 2017; de Souza and Nes 1968).

Procházková et al. (2017) reported sterol concentrations of up to 2.25 mg/L in water containing cyanobacterial blooms, highlighting the presence of these compounds in aquatic environments. The biosynthesis of isoprenoids, beginning with the building block isopentenyl diphosphate, leads to the production of various cyclic and acyclic compounds, including the C30 oxygenated isoprenoid oxidosqualene. Sterols are produced by a series of intracellular enzymatic processes in the isoprenoid biosynthesis pathway (Fabris et al. 2014; Volkman 2003). As integral components of cell membranes, sterols play pivotal roles in numerous biochemical processes in eukaryotic cells, including the production of steroid hormones (Martin-Creuzburg And von Elert 2009). From a nutritional perspective, sterols are valuable dietary sources for aquacultured organisms, contributing to their health and growth (Cardozo et al. 2007). Structurally, sterols confer membrane stability and integrity due to their rigid fused ring system, which strengthens the membrane architecture. Studies have demonstrated that several cyanobacterial species actively participate in sterol biosynthesis, facilitated by enzymes such as D24-sterol C-methyltransferase, sterol-C-5-desaturase, C-4 methyl

sterol oxidase, and sterol-C-methyltransferase (Kaneko and Tabata 1997; Nes 2011). Although some cyanobacterial lineages have been shown to include genes for sterol biosynthesis, there is currently no experimental proof that sterol production occurs, suggesting that sterol biosynthesis is not a common trait of cyanobacteria.

Fatty acids

Fatty acids are among the most critical metabolites due to their central role in metabolic processes (Gago et al. 2011; Leão et al. 2023; Maltsev And Maltseva 2021). In cyanobacteria, polyunsaturated fatty acids contribute to membrane lipid composition and physiological adaptation, whereas in aquatic food webs these fatty acids function as essential nutrients for higher organisms (Hernando et al. 2022; Shiels et al. 2022; Djuricic And Calder 2021). As the primary constituents of lipids, fatty acids, and alcohols give rise to a diverse array of fats, phospholipids, glycolipids, and waxes, with structural variation depending on their arrangement. Long-chain C20 polyunsaturated fatty acids, such as arachidonic and eicosapentaenoic acids, have only been identified in few cyanobacterial species, but numerous cyanobacteria synthesize C18 fatty acids, such as linoleic and linolenic acids (Lopes et al. 2023; Singh et al. 2002). A notable study demonstrated that fatty acids present in the cell lysate of *Microcystis* effectively inhibited ion transport in the gills of *Oreochromis mossambicus* (Bury et al. 1998).

In cyanobacteria, fatty acid biosynthesis is driven by fatty acid synthase (FAS), which operates through the use of acyl carrier proteins (ACPs) as key co-factors (Froehlich et al. 1990; Liu et al. 2011; Kaczmarzyk And Fulda 2010; Kahn et al. 2023) (Fig. 3). This type II FAS complex produces fatty acids by utilizing freely dissociable ACPs, which subsequently serve as direct substrates for acyltransferases, facilitating the incorporation of fatty acids into membrane lipids (Elsheikh And Eltanahy 2024; Frentzen et al. 1983; Kaczmarzyk And Fulda 2010).

Research on the biological activities of fatty acids has demonstrated their potential as anticancer, antibiotic, antifungal, and antiviral agents (Burja et al. 2001; El-Baz et al. 2013; Arokiaarajan et al. 2022; de Sousa Victor et al. 2020; Liao et al. 2023; Patel et al. 2023; Tonon et al. 2024). Among the various types of fatty acids, polyunsaturated fatty acids (PUFAs) are of particular interest due to their well-documented health benefits and growing demand in the global market (Khan et al. 2024; Steinhoff et al. 2014; Nagarajan et al. 2021). PUFAs, characterized by the presence of two or more methylene-interrupted double bonds, are highly valued from a nutraceutical perspective, given their association with numerous positive health outcomes. These fatty acids also exhibit significant biological activity, making them highly relevant for therapeutic interventions in cardiovascular diseases and obesity-related conditions (Cardozo et al. 2007; Gogna et al. 2023; Lee et al. 2016). Furthermore, PUFAs play a crucial role in regulating essential cellular

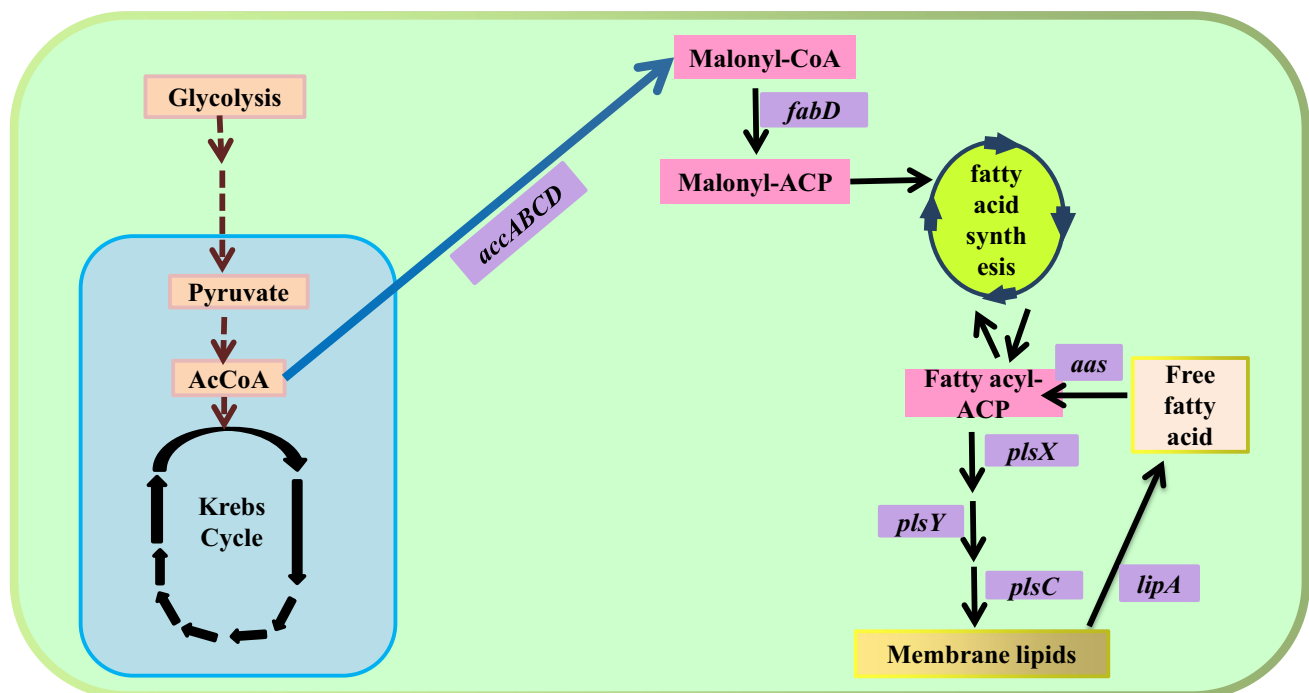


Fig. 3 The schematic diagram represents the biosynthesis pathway of fatty acid in cyanobacteria. The enzymes involved in the process encoded by different genes such as *accABCD*, multi-subunit acetyl-

CoA carboxylase; *fabD*, malonyl coenzyme A-acyl carrier protein transacylase; *lipA*, lipolytic enzyme; *plsX*, *plsY*, *plsC*, putative phosphate acyl-transferases

processes such as oxygen and electron transport, membrane fluidity, and thermal adaptation (Cardozo et al. 2007; Funk 2001). The multifaceted health benefits and biological functions of PUFAs highlight their increasing importance in both medical and commercial applications.

Photoprotective/UV-screening compounds

Numerous photoprotective secondary metabolites that cyanobacteria produce aid in reducing UV radiation-induced damage (Balskus And Walsh 2010; Gao et al. 2021; Mishra et al. 2024; Pathak et al. 2020). The two most significant of these are scytonemin and mycosporine-like amino acids (MAAs) (Kumar et al. 2021). The biosynthesis of these compounds is influenced by a variety of environmental stimuli, such as variations in light intensity, UV radiation, nutrient limitations, and other forms of stress (Görünmek et al. 2024; Rastogi et al. 2010). Photoprotective substances like scytonemin and MAAs work in conjugation to protect cyanobacterial cells from UV light. By absorbing UV rays and releasing surplus energy as heat, MAAs stop reactive oxygen species from forming (Oren And Gunde-Cimerman 2007; Whitehead And Hedges 2005; Rastogi et al. 2010). The extracellular sheath contains a pigment called scytonemin, which acts as a passive UV-screening agent to lessen UV penetration into cells (Singh et al. 2008). When combined, these substances offer protection against radiation-induced damage both within and outside of cells.

MAAs, in particular, are small, hydrophilic, colourless intracellular molecules with potential to dissipate excess energy as heat, thus preventing the formation of toxic oxygen radicals (Conde et al. 2000; Groniger And Hader 2000; Kageyama And Waditee-Sirisattha 2018; Oren And Gunde-Cimerman 2007; Whitehead And Hedges 2005). These protective compounds play a crucial role in mitigating the damaging effects of radiation stress, and there is growing evidence that MAAs can also extend photoprotection to primary and secondary consumers when ingested (Bhatia et al. 2011; Helbling et al. 2002; Singh et al. 2021; Quintero-Pacheco et al. 2024). The photoprotective role of MAAs highlights their ecological significance and potential applications in biotechnology and environmental protection. A discussion of cyanobacterial photoprotective compound categories and their roles have been elaborately described under different subheadings.

Mycosporine-like Amino Acids (MAAs)

MAAs are photoprotective substances produced by some cyanobacteria, algae, and other microbes. The cyclohexenone or cyclohexenimine chromophore is attached to amino

acids or amino alcohol substituents to form MAAs, which are small, water-soluble secondary metabolites (Bhatia et al. 2011; Nowruziet al. 2020; Singh et al. 2008). According to several studies, MAAs are produced by a variety of cyanobacteria, rhodophyta, and other groups of microalgae (Carreto And Carignan 2011; Rastogi et al. 2015; Rastogi And Madamwar 2016; Sinha et al. 2007). Numerous workers reported that these compounds have UV-range absorption maxima (Bhatia et al. 2011; Kannaujiya et al. 2014; Takano et al. 1979).

In cyanobacteria, MAA biosynthesis is carried out *via* enzymatic pathways connected to the pentose phosphate and shikimate pathways. Sedoheptulose-7-phosphate has been found to be a crucial precursor in the formation of the MAA chromophore (Rastogi et al. 2015). 3-Dehydroquinate is a precursor for molecules involved in the biosynthesis of MAAs in cyanobacteria and an intermediate in the shikimate pathway (Favre-Bonvin et al. 1987). It is used in skin protection products like sunscreen and other non-biological materials like photo stabilizing additives in varnish, paint, and plastics. MAAs, which elicit protective reactions, are believed to have evolved from cyanobacteria and algae (Klisch and Häder 2008; Nakamura et al. 1982). Biosynthesis of MAAs involves two key steps: first, reduction of carboxylic group of 3-dehydroquinate; second, methylation of hydroxyl group at C4, followed by attachment of one amino acid (forming mycosporine-glycine) or two amino acids or amino alcohols (Klisch and Häder 2008) (Fig. 4).

Strong evidence in favour of MAAs as photoprotective compounds comes from their high molar co-efficients UV-absorption maxima ranging from 310nm to 362nm, photostability in fresh and salt water in the presence of photosensitizers, and resistance to primarily abiotic stressors like pH, temperature, various solvents, and UVR (Nowruziet al. 2020; Shahidi And Zhong 2010; Whitehead And Hedges 2005). The species and the localization of the pigments play a major role in the protection against UV-B toxicity offered by MAAs (Fig. 5). Several cyanobacteria with cytoplasmic MAAs have been found to provide significant, but limited, protection (Fig. 5). Garcia-Pichel and Castenholz, (1993) found that only 10–26% of the photons are absorbed by the pigment when MAAs are located in the cytoplasm. Since MAAs are found in the extracellular glycocalyx in the *Nostoc commune*, it is believed that they play a significant role in photoprotection. According to Böhm et al. (1995), two out of every three photons that enter a cell are absorbed by pigments found in cell membranes or targets. Two UVA/B-absorbing pigments in *N. commune* were found to be present in colonies when exposed to high solar radiation (Ferroni et al. 2010; Scherer et al. 1988). These pigments, with absorption maxima at 312 and 335 nm, include one of the earliest identified mycosporines found within the

Fig. 4 The diagram shows the different routes for biosynthesis of MAAs in cyanobacteria

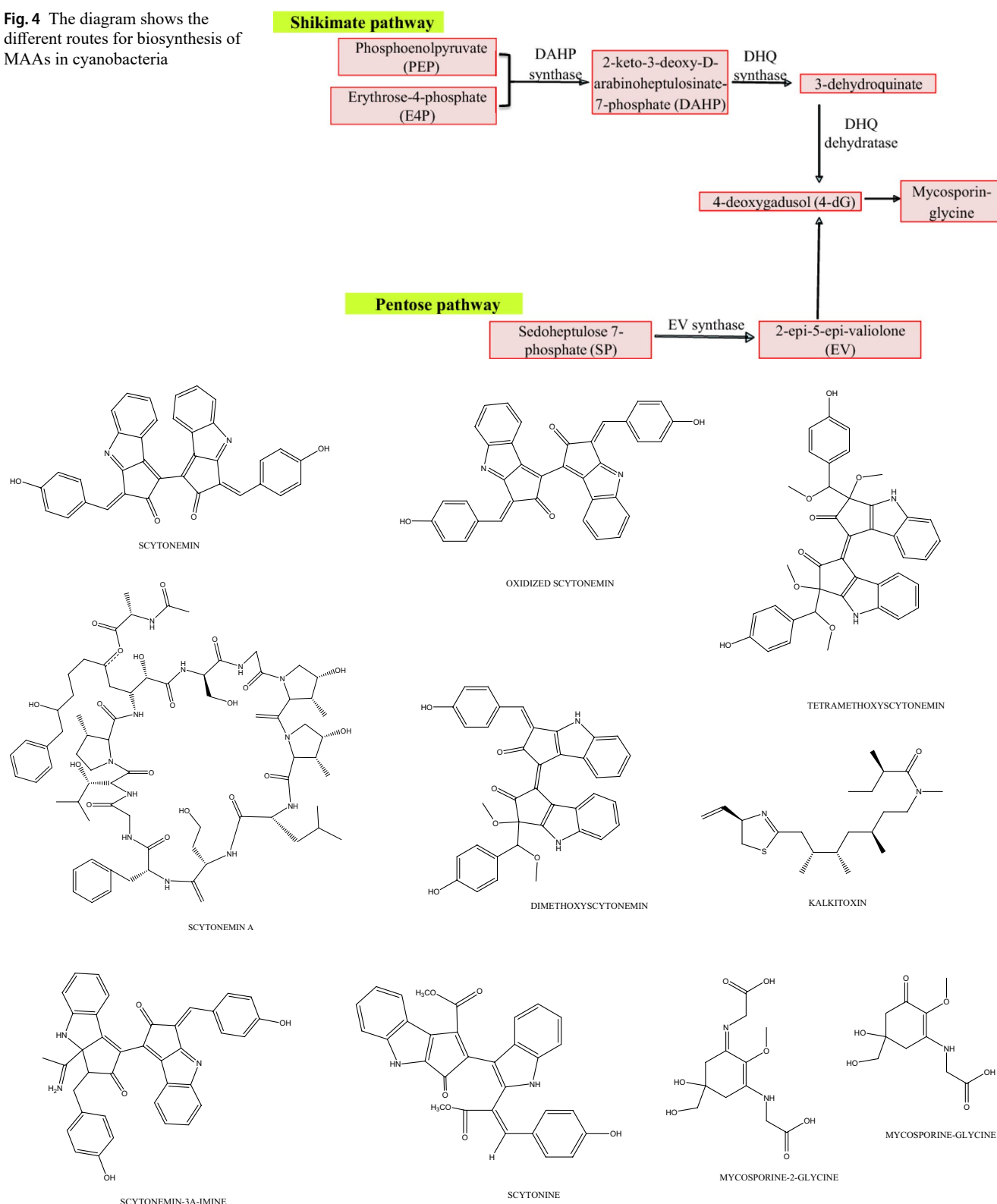


Fig. 5 Cyanobacterial secondary metabolites from several metabolite classes are displayed with their representative chemical structures. The structures show the variety of chemical structure cyanometabolites,

which includes terpenoids, alkaloids, polyketides, peptides, and other bioactive substances

extracellular glycan, where it is covalently bound to oligosaccharides (Böhm et al. 1995; Hill et al. 1994). These compounds effectively dissipate absorbed UV radiation as heat, preventing the formation of harmful reactive oxygen species (ROS) (Conde et al. 2000). Furthermore, MAAs not only protect their primary producers but also extend photoprotection to primary and secondary consumers within the food chain, shielding them from UVR (Helbling et al. 2002). The synthesis of MAAs is intricately linked to the photosynthetic process. This was demonstrated by the inhibition of MAA biosynthesis in the alga *Alexandrium excavatum* when treated with 3-(3,4-dichlorophenyl)-1,1-dimethylurea (DCMU), a photosynthesis inhibitor, suggesting a direct connection between photosynthesis and MAA production.

Scytonemin

Scytonemin, a photoprotective compound with a molecular weight of 544 Da, is a dimeric structure composed of phenolic and indolic subunits (Fig. 5). Initially identified as a yellowish-brown, lipid-soluble pigment within the exopolysaccharide sheath of various terrestrial cyanobacteria (Garcia-Pichel And Castenholz 1991; Nowruziet al. 2020; Pathak et al. 2020; Rastogi et al. 2015; Rothrock and Garcia-Pichel 2005; Wada et al. 2013). This compound exists in three distinct forms: the commonly found oxidized green form, a reduced red variant known as fuscorhodin, and an oxidized yellow variant termed fuscochlorin (Garcia-Pichel And Castenholz 1991; Wada et al. 2013). Recent investigations have also isolated additional scytonemin derivatives, including dimethoxy scytonemin, tetramethoxy scytonemin, and scytonemin pigments, from organic extracts of *Scytonema* (Bultel-Poncé et al. 2004; Grant And Louda 2013; Rastogi et al. 2014). The absorbance spectrum of scytonemin indicates that the live-cell variant exhibits peak absorption at 370 nm, while purified scytonemin absorbs most effectively at 386nm. Notably, it also demonstrates substantial absorbance at 252, 278, and 300nm, which likely contributes to its protective role in helping cyanobacteria to withstand the damaging effect of UV radiation. Several researches have shown that scytonemin alone can significantly mitigate the risks associated with exposure to the most harmful UV-C radiation (Dillon And Castenholz 1999; Rastogi et al. 2013).

Scytonemin has been shown to significantly mitigate both the photobleaching of chlorophyll *a* and the inhibition of photosynthesis induced by UV-A radiation (Cockell And Knowland 1999; Gao and Garcia-Pichel 2011; Nowruziet al. 2020). This photoprotective capability is exemplified in *Chlorogloeopsis* sp., where scytonemin functions effectively as a UV sunscreen (Garcia-Pichel et al. 1992; Portwich and Garcia-Pichel 2003). Unlike other UV protective

mechanisms, which rely on the active repair of damaged cellular components and may fail after extended periods of physiological inactivity, scytonemin exhibits stability in response to various stressors, including UV radiation and temperature fluctuations. This stability allows scytonemin to perform its screening activity without necessitating additional metabolic responses. Moreover, the superior UV screening potential of scytonemin suggests its potential application as a natural sunscreen in human cosmetics, highlighting its dual role as both a protective agent in cyanobacterial physiology and a promising ingredient for photoprotection in cosmetic formulations (Nowruziet al. 2020; Rastogi et al. 2010). Figure 5 represents different types of scytonemin reported in cyanobacteria.

Halogenated substances

Halogenated compounds, often mistakenly believed to be solely anthropogenic, have been widely isolated from marine algae, particularly brown and red algae. These compounds, including acetogenins, phenols, terpenes, indoles, fatty acids, and volatile halogenated substances such as dibromomethane, chloroform, and bromoform (Butler And Carter-Franklin 2004), are also prevalent across cyanobacteria and various other group of algae. Their pharmacological significance is underscored by their diverse biological activities, which include antiproliferative, antifungal, antibacterial, antiviral, antifeedant, antifouling, anti-inflammatory, cytotoxic, ichthyotoxic, insecticidal, and antitumoral properties (Cabrita et al. 2010; Vairappan et al. 2001). Additionally, bioactive peptides like aeruginosin and cyanopeptolin, recognized as potent protease inhibitors, have been extensively documented in numerous cyanobacteria, offering promising applications in both agrochemistry and pharmaceutical development (Silva-Stenico et al. 2011).

Phytohormones

Auxins such as indole-3-acetic acid (IAA), gibberellins (GA), cytokinins (CKs), and ethylene (ET), key regulators of plant growth and development, have been extensively documented to accumulate and be released by various cyanobacterial and algal genera (Gayathri et al. 2015; Hussain And Hasnain 2012; Stirk et al. 2002). Symbiotic cyanobacteria, notably through the excretion of phytohormones such as IAA and CKs, significantly contribute to host plant growth and development (Hussain And Hasnain 2011). A pivotal study by Hussain et al (2013) revealed that endophytic *Nostoc* strains produce comparable levels of IAA and CKs in the root cells of both rice and wheat. The *ipt* (isopenentenyltransferase) gene catalyzes the transfer of isopenentenyl groups to adenine nucleotides, which directly contributes

to the production of cytokines. Indole-3-acetic acid (IAA) biosynthesis does not directly include the *ipt* gene, despite the fact that cytokinins can affect physiological processes related to auxins. Gene knockout experiments resulted in a marked reduction of CKs and IAA due to the deactivation of the *ipt* gene, which occurred *via* homologous recombination. This genetic modification led to a significant decline in zeatin production, followed by a pronounced reduction in the growth of the mutant strain. Additionally, numerous plant growth regulators (PGRs), including amino acids, sugars, and vitamins, have been isolated from various cyanobacterial and algal genera, which may further enhance the growth of vascular plants (Karthikeyan et al. 2007; Misra And Kaushik 1989).

Cyanotoxins

Marine and freshwater cyanobacteria as well as algae are known to produce array of toxins, classified as secondary metabolites. Certain cyanobacterial species produce biologically active secondary metabolites called cyanotoxins, which can be harmful to aquatic habitats, humans, and animals. These substances are usually categorized as hepatotoxins, neurotoxins, cytotoxins, and dermatotoxins based on their main physiological targets. The synthesis of cyanotoxins is frequently linked to toxic algal blooms that are brought on by eutrophication, temperature fluctuations, and nutritional imbalance. From an ecological point of view, cyanotoxins might serve as competitive agents or chemical defense molecules, but from a public health standpoint, they are important environmental pollutants that have an impact on aquatic food webs and drinking water safety. Hepatotoxins represent one of the most extensively studied groups of cyanotoxins due to their severe effects on liver function. The most prominent hepatotoxins, including microcystins and nodularins, primarily target protein phosphatases (PP1 and PP2A), leading to disruption of cytoskeletal organization, oxidative stress, and hepatocyte damage. Because these toxins can bioaccumulate in aquatic food chains and contaminate drinking water sources, they are of particular concern in environmental toxicology and public health monitoring.

The formation of harmful algal blooms, driven by nutrient overloads such as nitrogen and phosphorus, significantly degrades water quality and leads to the release of various toxins (Codd et al. 2005). Among the five primary groups of cyanobacterial and algal toxins, neurotoxins such as anatoxins and saxitoxins are particularly notable. Other toxins of serious concern, due to their adverse effects on human health, include cytotoxins (e.g., cylindrospermopsin), hepatotoxins (e.g., nodularin and microcystins), dermatotoxins, and irritant toxins such as lipopolysaccharides (Gácsi et al. 2009; Wiegand And Pflugmacher 2005). These toxic

compounds pose significant threats to both ecological system and public health, warranting focused attention on their regulation and mitigation.

Cyanobacteria such as *Anabaena*, *Microcystis*, *Nostoc*, and *Oscillatoria* have been identified as primary sources of potent toxins, including cylindrospermopsin, microcystins, and saxitoxins (Codd et al. 2005; Lalić 2024; Lei et al. 2024). Another notable toxin, cylindrospermopsin, an alkaloid is produced by *Aphanizomenon ovalisporum* strains in Australia and Israel (Valério et al. 2005). Toxins produced by *Anabaena*, *Umezakia natans*, and *Cylindrospermopsis raciborskii*, found in Japan, Australia, and the United States, respectively, further exemplify the global distribution of these harmful compounds (Neumann et al. 2007; Törökne et al. 2004). Among cyanobacterial neurotoxins, three major classes are particularly concerning: (i) anatoxin-a, the first identified and highly potent cyanotoxin (Koskinen And Rapoport 1985); (ii) saxitoxin, produced by *Anabaena circinalis* in Australia and *Aphanizomenon flos-aquae* in North America, which has been linked to widespread animal fatalities (Mahmood And Carmichael 1986); and (iii) anatoxin-a(s), a potent irreversible acetylcholinesterase inhibitor (Devic et al. 2002). These toxins render water unsafe for consumption and have detrimental effects on herbivorous zooplankton populations (Hansson et al. 2007), posing significant public health risks (Stewart et al. 2006). Of the five major toxin groups, hepatotoxins and neurotoxins are the most dangerous to humans and animals due to their bioaccumulation in the liver and kidneys, respectively (Wiegand And Pflugmacher 2005).

Cyanobacterial toxins use specific molecular targets to achieve their biological effects. The inhibition of serine/threonine protein phosphatases such as PP1 and PP2A by hepatotoxic microcystins and nodularins results in oxidative stress, hepatocyte injury, and cytoskeletal disruption (Compos And Vasconcelos, 2010). While anatoxin-a functions as a strong agonist of nicotinic acetylcholine receptors, resulting in neuromuscular dysfunction, neurotoxins like saxitoxin interfere with the transmission of neural signals by blocking voltage-gated sodium channels (Florczyk et al. 2014; Grishin 2005). Cylindrospermopsin mainly causes oxidative stress and inhibits the production of proteins in the tissues of the liver and kidneys (Sieroslawska And Rymuszka 2015). These mechanistic revelations emphasize the ecological function of cyanotoxins in chemical defense as well as its applicability to environmental monitoring and toxicity.

Siderophore

The ability to chelate iron is a common property in both freshwater and marine cyanobacteria, and some of these

chelators also operate as siderophores (siderous = iron; phorus = bearer) (Kerry et al. 2008; Rastogi and Sinha 2009). In situations of iron stress, bacteria release siderophores, which are low molecular weight, high affinity Fe(III)-binding ligands that scavenge and transport iron. It has also been demonstrated that, in iron-limiting condition, cyanobacteria can make siderophores, which are low-molecular-weight (400–1200 Da) ferric-specific chelators (Wilhelm And Trick 1994). When there is an iron shortage, siderophores attach to Fe^{3+} to create a ferrisiderophore complex, which makes it easier for ferric ions to enter cells (Matzanke 1991). *Anabaena* PCC 7120 was the first to be identified as the source of Schizokinen, a citrate derivative siderophore (Goldman et al. 1983). *Synechococcus* PCC 7002 has yielded synchobactins, which are thought to be schizokinen derivatives. They function as siderophores and have one of the hydroxamic acids substituted with a long fatty acid (Ito And Butler 2005). The biofilm-forming cyanobacterium, *Anabaena cylindrica* 1403-2a was the source of two complex siderophores, anachelin H with a terminal salicylamide and anachelin 1, ended by an oxazoline ring (Beiderbeck et al. 2000). Siderophores are particularly intriguing as potential medications for use in the fields of biomaterials and biosensors (Gademann et al. 2009), as well as in the treatment of iron overload and acute and chronic metal toxicity (Neilands 1981).

Ribosomally synthesized and post-translationally modified peptides

Ribosomally synthesized and post-translationally modified peptides (RiPPs) represent a diverse class of secondary metabolites that undergo significant structural modifications after ribosomal synthesis. These post-translational modifications (PTMs) include leader peptide hydrolysis, cyclization, and the formation of disulfide bonds, among others. The biosynthetic gene clusters responsible for RiPP synthesis typically encode a short precursor peptide, characterized by an N-terminal leader sequence and a C-terminal core sequence, alongside PTM enzymes that are essential for structural diversification (Arnison et al. 2013; Montalbán-López et al. 2021). PTM enzymes act on linear precursor peptides, introducing various modifications that yield functionally diverse and structurally complex mature peptides. RiPPs were previously categorized into subgroups such as lanthipeptides, lasso peptides, microviridins, cyanobactins, and microcins. However, they are now unified under the RiPP classification due to their shared biosynthetic mechanisms. These compounds exhibit range of biological activities, including protease inhibition, cytotoxicity, signalling, anticancer properties, and even resistance to human

immunodeficiency virus (HIV) (Arnison et al. 2013). For instance, microviridin, initially isolated from *Microcystis viridis*, acts as a potent serine protease inhibitor, while patelamide A, produced by *Prochloron didemni*, demonstrates its cytotoxic effects (Schmidt et al. 2005). These bioactive properties highlight the pharmaceutical potential of RiPPs, underscoring their importance in drug discovery and therapeutic development. Enzymatic tailoring produces structurally restricted compounds with high target selectivity from ribosomally produced and RiPPs. In contrast to cyanobactins, which demonstrate cytotoxic or antibacterial activity by membrane contact or enzyme inhibition (Khan And Phan 2025), microviridins serve as serine protease inhibitors by attaching to the catalytic site of target enzymes (Weiz et al. 2011). Following enzymatic cyclization, methylation, and oxidation events that influence biological activity, the biosynthesis of RiPPs entails precursor peptides encoded in biosynthetic gene clusters (Han And Won 2024).

Cyclic peptide

Cyclic peptides, distinctive class of natural products that are known for their structural diversity and extensive pharmacological applications. Over the past two decades, numerous cyclic peptides have been isolated from marine cyanobacteria, revealing novel scaffolds, halogenated molecules, and intricate spatial configurations (Tan 2007). These discoveries highlight the underexplored chemical diversity of cyanobacteria as a rich source of bioactive compounds. Among the most extensively studied hepatotoxins are microcystins and nodularins (Ferrão-Filho And Kozłowski-Suzuki 2011). Microcystins, originally isolated from the cyanobacterium, *Microcystis aeruginosa*, have also been detected in other cyanobacterial genera (Buratti et al. 2017). Microcystins are characterized by a cyclic heptapeptide structure, which includes seven amino acids. A hallmark of their structure is the presence of the unique 3-amino-9-methoxy-2,6,8-trimethyl-10-phenyldeca-4,6-dienoic acid moiety (ADDA) which plays a critical role in their bioactivity. These toxins have significant implications for human and animal health due to their potent hepatotoxic properties. The structural complexity and bioactivity of cyanobacterial cyclic peptides underscore their significance in drug discovery, environmental monitoring, and toxicology, making them a crucial subject of study in natural product research.

The genera responsible for producing microcystins include *Microcystis*, *Planktothrix*, and *Dolichospermum*, while the nodularin (cyanotoxin) is predominantly produced by *Nodularia* sp. Cryptophycin-1, initially separated from the cyanobacterium, *Nostoc* sp. for its antifungal properties, was later re-isolated in 1992 from another *Nostoc* sp. (GSV 224) as an anticancer agent (Liang et al. 2005). Numerous

cryptophycin analogues have been discovered, both naturally and through chemical synthesis. Notably, cryptophycin-52 (LY355073) advanced to phase 2 clinical trials for treating platinum-resistant ovarian cancer. Apratoxins A-G, a class of cyclic lipopeptides isolated from the marine cyanobacterium *Lyngbya* sp., exhibit potent cytotoxic activity (Gutiérrez et al. 2008; Luesch et al. 2001). Apratoxin A demonstrated high cytotoxicity in vitro, with IC₅₀ values of 0.52 nM against KB cells and 0.36 nM against Lovo cancer cells. Key structural features contributing to its cytotoxicity include hydroxyl groups and N-methylated isoleucine at C-35. Comparatively, apratoxin analogues B and C showed reduced cytotoxicity. Apratoxin D exhibited an IC₅₀ value of 2.6 nM against H-460 human lung cancer cells, which matched the potency of apratoxin A. Apratoxins F and G displayed cytotoxic activity against H-460 cells with IC₅₀ values of 2 nM and 14 nM, respectively. Further research on the cyanobacterium, *Moorea bouillonii*, collected from the northern coast of New Britain, Papua New Guinea, led to the discovery of a novel cytotoxic cyclic depsipeptide known as bouillonamide (Tan et al. 2013).

Cyanobacterial cyclic peptides have a variety of pharmacological actions that are mediated by certain molecular targets. Protein phosphatases namely PP1 and PP2A are inhibited by microcystins (MacKintosh et al. 1990), whereas cryptophycins bind to tubulin and prevent mitosis, disrupting microtubule assembly (Smith and Zhang 1996). By preventing co-translational translocation in the endoplasmic reticulum, apratoxins disrupt the secretory system, lowering growth factor signaling and triggering death (Qamar et al. 2021). Cyclic peptides strong binding affinity and biological potency are a result of their structural rigidity and conformational stability (Buchanan et al. 2025).

Acyclic peptides

Many cyanobacterial peptides that lack the cyclic backbone structure and are linear peptide metabolites called as acyclic peptides. These substances frequently have strong biological activity despite having a simpler structural arrangement because they contain unusual amino acids and post-translational modifications. The potential of cyanobacteria-derived acyclic peptides as lead molecules in the development of anticancer drugs has been highlighted by their significant cytotoxic and antiproliferative characteristics. Dolastatin 10, a potent cytotoxic peptide composed of rare amino acids, was originally discovered in the sea hare *Dolabella auricularia* (Pettit et al. 1987). Subsequent studies revealed its production by the marine cyanobacterium, *Symploca* sp. (Luesch et al. 2001). The synthetic analogue of dolastatin 10, monomethyl auristatin E (MMAE), was conjugated with a chimeric human-murine IgG1 antibody to develop the

antibody–drug conjugate brentuximab vedotin (Adcetris®) (Solanki 2020). This drug demonstrated significant efficacy against human colon adenocarcinoma and was granted Food and Drug Administration (FDA) US approval in 2011, followed by European approval in 2015 for the treatment of lymphomas (Rangel And Falkenberg 2015).

The urgent need for cyanobacterial metabolites as antiviral agents and current state of antiviral drugs

Cyanobacterial raw extracts and their secondary metabolites have demonstrated significant potential in addressing various human health challenges. A range of techniques, including in silico, in vitro, and in vivo methods have been employed to identify and characterize their pharmacological properties. Over the past five decades, the global population has faced a considerable burden due to the absence of vaccines and effective treatments for several viral infections, including hepatitis C virus (HCV) and HIV. In particular, no specific drug candidates are currently available for treating certain viral infections, such as respiratory tract viruses like Herpes Simplex Viruses (HSV-1 and HSV-2). In the past 16 years, the United Nations Millennium Development Goals (UNMD) initiative has allocated financial resources to reduce the incidence of infectious diseases in low-income nations (Dye and O'Garra 2014). However, many existing antiviral drugs on the market are associated with adverse side effects, particularly when used to treat individuals infected with these viruses. This challenge has prompted researchers to explore novel natural antiviral solutions. Consequently, scientists are utilizing diverse approaches to extract bioactive compounds from cyanobacteria, aiming to discover promising new antiviral candidates (Ahmadi et al. 2015).

Over the past two decades, researchers have employed various solvents such as water, methanol, and ethanol to extract secondary metabolites from cyanobacteria, aiming to discover novel antiviral compounds. Studies have indicated that the efficacy of these antiviral compounds varies depending on the research focus and specific pharmacological applications (Ahmadi et al. 2015; Sanmukh et al. 2014). Supporting this, Lau et al. (1993) demonstrated that cyanobacterial strains possess potent viral inhibitory activities, particularly against HIV and the avian myeloblastosis virus, by inhibiting reverse transcriptase activity (Tables 1, 2, and 3).

Table 1 Metabolites synthesized by different groups of cyanobacteria and algae (adopted from Singh et al. 2017)

Metabolites	Algae/cyanobacteria	Uses	References
Phenolic compounds	<i>Arthrospira platensis</i> <i>Nostoc muscorum</i> , <i>Phormidium foveolarum</i> , <i>Spirulina platensis</i>	Prevents vascular damage as well as cardiovascular diseases progression Provides defence by scavenging the free radicals Acts as a UV-B screening compound Provide resistance to plants against pathogens, pests, and diseases	Rice-Evans et al. 1997; Singh et al. 2003; La Camera et al. 2004; Vogt 2010; Ferrari et al. 2015; Kumar et al. 2016;
Fatty acids	<i>Synechocystis</i> sp., <i>Anabaena</i> , <i>Nostoc</i> , <i>Spirulina</i> <i>Phaeodactylum tricornutum</i> , <i>P. lutheri</i> , <i>Nostoc commune</i>	Provides chemical defence as it is toxic to grazers Lipid extracts may be used as herbal medicine, to treat against cancer, viruses, burns, and chronic fatigue	Anupama And Ravindra 2000; Jüttner 2001; Rasmussen et al. 2008; Guedes et al. 2011; Hellier et al. 2013
Terpenoids	<i>Synechocystis</i> sp.	Provides chemical defence against herbivory, fragrances, and flavours	Kirby And Keasling 2009; Bentley et al. 2013
Mycosporineglycine, Porphyra-334, and Shionine	<i>Anabaena doliolum</i> , <i>Scytonema javanicum</i>	Acts as osmolytes and improves antioxidant status, which subsequently lowers the level of ROS	Oren and Gunde-Cimerman 2007; Klisch and Häder 2008; Singh et al. 2010
Carotenoids, β -Carotene, Lutein, Zeaxanthin, Cryptoxanthin, α -carotene, Lycopene	<i>Nostoc muscorum</i> , <i>Phormidium foveolarum</i> , <i>Spirulina platensis</i>	Improve antioxidant status of the organisms Use in food industry Use to prevent cancer	Cardozo et al. 2007; Prasanna et al. 2010; Kumar et al. 2016
Scytonemin	<i>Scytonema</i>	Pharmacological potential and used as anti-inflammatory and antiproliferative agent	Stevenson et al. 2002
Carragenans, Agar, and Lectins	<i>Chondrus ocellatus</i> *	Widely used as antitumor, antiviral, anticoagulant, and immunomodulation agent	Marinho-Soriano And Bourret 2003; Jepson et al. 2004; Cardozo et al. 2007
Halogenated compounds	<i>Synechococcus elongates</i> PCC7942, <i>Cylindrospermopsis raciborskii</i> 339-T3, <i>Fischerella</i> , <i>Microcystis aeruginosa</i> NPCD-1, <i>Microcystis panniformis</i> SCP702	Exhibit antiviral, antifungal, antifouling, antiproliferative, antibacterial, anti-inflammatory activity Show cytotoxic, antifeedant, insecticidal, and ichthyotoxic responses	Blunt et al. 2009; Rastogi and Sinha 2009; Silva-Stenico et al. 2011
Phytohormones	<i>Anabaena vaginicola</i> , <i>Nostoc calcicola</i>	Enhanced the production of defence enzymes and provide protection against different stresses	Takahama et al. 2003; Tarakhovskaya et al. 2007; Hashtroudi et al. 2013
Toxins	<i>Microcystis</i> sp., <i>Anabaena</i> sp., <i>Oscillatoria</i> sp., <i>Anabaenopsis</i> sp., <i>Nostoc</i> sp., <i>Hapalosiphon</i> sp., <i>Lyngbya polychroa</i>	Used for development of bioicides that serve as antibiotics, anticancerous, and anti-inflammatory agents with relevant to pharmaceutical activities	Burja et al. 2001; Biondi et al. 2004; Cardozo et al. 2007; Gunasekera et al. 2008

Organisms that are eukaryotic algae are represent with “*” mark

Table 2 Various phytohormones produced by cyanobacteria and algae and their potential implications in microbial biotechnology (adopted from Singh et al. 2017)

Phytohormones	Cyanobacteria/algae	Relevant findings in microalgae	Potential implication for microalgal biotechnology	References
Abscisic acid (ABA)	<i>Anabaena variabilis</i> , <i>Ascomyllum nodosum</i> *, <i>Chlamydomonas reinhardtii</i> *, <i>Chlorella minutissima</i> *, <i>Chondracanthus</i> sp., <i>Coscinodiscus granii</i> *, <i>Draparnaldia mutabilis</i> *, <i>Dunaliella</i> sp., <i>Gelidium</i> sp., <i>Gracilaria</i> sp., <i>Gracilariopsis</i> sp., <i>Hypnea</i> sp., <i>Nannochloropsis oceanica</i> *, <i>Nostoc muscorum</i> , <i>Porphyr</i> a sp., <i>Trichormus variabilis</i> , <i>Synechococcus leopoliensis</i>	Exogenous ABA decreases growth rate in <i>Nannochloropsis oceanica</i> * and <i>Coscinodiscus granii</i> * Exogenous ABA improves stress tolerance to dehydration in <i>Haematococcus pluvialis</i> *, higher salinity in <i>Dunaliella</i> sp.,* and <i>Chlamydomonas reinhardtii</i> *; nitrogen deprivation in <i>Nannochloropsis oceanica</i> *; osmotic stress in <i>Chlamydomonas reinhardtii</i> *	Improvement in stress tolerance	Boyer And Dougherty 1988; Hirsch et al. 1989; Kentzer and Mazur 1991; Zahradnickova et al. 1991; Maršálek et al. 1992; Tomimaga et al. 1993; Kobayashi et al. 1997; Yoshida et al. 2003; Hartung 2010; Yokoya et al. 2010; Lu et al. 2014; Stirk et al. 2014
Auxins	<i>Anabaena</i> sp., <i>Chlorella minutissima</i> *, <i>Chlorella pyrenoidosa</i> *, <i>Chondracanthus</i> sp., <i>Chroococcidiopsis</i> sp., <i>Ectocarpus siliculosus</i> *, <i>Gelidium</i> sp., <i>Gracilaria</i> sp., <i>Gracilariopsis</i> sp., <i>Hypnea</i> sp., <i>Nostoc</i> sp., <i>Oscillatoria</i> sp., <i>Phormidium</i> sp., <i>Porphyr</i> a sp., <i>Prionitis lanceolata</i> *, <i>Scenedesmus armatus</i> *, <i>Synechocystis</i> sp.	Exogenous indole-3-acetic acid (IAA) improves growth rate in <i>C. reinhardtii</i> *, <i>Chlorella sorokiniana</i> *, <i>Chlorella vulgaris</i> *, <i>Haematococcus pluvialis</i> *, <i>Nostoc</i> sp., <i>Phaeodactylum triicornutum</i> *, <i>Chlorella pyrenoidosa</i> *, and <i>Pleurochrysis carterae</i> *, and oil content in <i>Chlamydomonas reinhardtii</i> *, <i>Haematococcus pluvialis</i> *, <i>Phaeodactylum triicornutum</i> *	Elevation of microalgal growth rate, biomass production, oil content, and stress tolerance	Ashen et al. 1999; Mazur et al. 2001; Sergeeva et al. 2002; Bail et al. 2010; Hussain et al. 2013; Maor. 2012; Yokoya et al. 2010; Mazhar et al. 2013; Park et al. 2013; Piotrowska-Niczyporuk And Bajguz 2014; Stirk et al. 2014
Cytokinins (CK)	<i>Anabaena</i> sp., <i>Calothrix</i> sp., <i>Chlorella minutissima</i> *, <i>Chlorogloeopsis</i> sp., <i>Chondracanthus</i> sp., <i>Chroococcidiopsis</i> sp., <i>Ecklonia</i> sp., <i>Ecklonia maxima</i> *, <i>Gelidium</i> sp., <i>Gigartina clathrate</i> *, <i>Gracilaria</i> sp., <i>Gracilariopsis</i> sp., <i>Hypnea</i> sp., <i>Laminaria pallid</i> *, <i>Nannochloropsis oceanica</i> *, <i>Oscillatoria</i> sp., <i>Phormidium</i> sp., <i>Porphyr</i> a sp., <i>Rhodospirillum</i> sp., <i>Synechocystis</i> sp.	Exogenous CK improves cell cycle progression in <i>Nannochloropsis oceanica</i> *; growth rate in <i>C. reinhardtii</i> *, <i>Nannochloropsis oceanica</i> *, and oil content in <i>C. reinhardtii</i> *, <i>Haematococcus pluvialis</i> *, and <i>Phaeodactylum triicornutum</i> * Elevated temperature led to increase in CK contents in <i>Ecklonia maxima</i> * and <i>Macrocystis pyrifera</i> * ET take part in programmed cell death of microalgae in <i>Chlamydomonas reinhardtii</i> *	Elevation of microalgal growth rate, oil content, and stress tolerance	Jennings 1969; Tian et al. 2006; Tsavkelova et al. 2006; Hussain et al. 2010; Stirk et al. 2011; Park et al. 2013; Lu et al. 2014; Stirk et al. 2013, 2014
Ethylene (ET)	<i>Anabaena</i> sp., <i>Calothrix</i> sp., <i>Chlorella pyrenoidosa</i> *, <i>Cylindrocapsa</i> sp., <i>Ecklonia maxima</i> *, <i>Nostoc</i> sp., <i>Padina arborescent</i> sp., <i>Porphyr</i> a tenera*, <i>Scytonema</i> sp., <i>Synechococcus</i> sp.		Enhancement of microalgal growth rate as well as biomass productivity	Watanabe And Kondo 1976; Kreslavsky et al. 1997; Tsavkelova et al. 2006; Yordanova et al. 2010
Gibberellins (GA)	<i>Anabaenopsis</i> sp., <i>Chlamydomonas reinhardtii</i> *, <i>Chlorella</i> sp., <i>Cylindrocapsa</i> sp., <i>Eckloniaradiata</i> *, <i>Hypnea musciformis</i> *, <i>Nannochloropsis oceanica</i> *, <i>Phormidium foveolarum</i>	Exogenous GA stimulates astaxanthin biosynthesis in <i>Haematococcus pluvialis</i> * Exogenous GA improves growth rate in <i>C. reinhardtii</i> *	Increased algal growth rate and biomass productivity. Target chemical production	Jennings 1968; Gupta And Agarwal 1973; Tsavkelova et al. 2006; Park et al. 2013; Stirk et al. 2013; Voß et al. 2014

Organisms that are eukaryotic algae are represent with "*"mark

Table 3 An overview of cyanotoxins produced by cyanobacteria and algae and their potential impacts on other organisms (adopted from Singh et al. 2017)

Cyanotoxins	Source	Nature	Effect	References
Cylindrospermopsins (Hepatotoxic, nephrotoxic, and cytotoxic)	<i>Anabaena</i> , <i>Aphanizomenon ovalisporum</i> , <i>Umezakianatan</i> , <i>Raphidiopsis curvata</i> , and <i>Cylindrospermopsis raciborskii</i>	Guanidine alkaloid	Inhibitor of protein biosynthesis, glutathione synthesis, imitate cytochrome P-450, overexpression of DNA damage repair proteins genotoxic	Humpage et al. 2000; Neumann et al. 2007
Lyngbyatoxin	<i>Lyngbya majuscula</i> , <i>Oscillatoria</i> , and <i>Schizothrix</i>	Alkaloid	Causes erythema (dermatitis), blisters, and necrosis in mammals; potent tumor promoters	Fujiki et al. 1983; Taylor et al. 2014
Dermatotoxins	<i>Lyngbya</i> , <i>Schizothrix</i> , <i>Gracilaria coronopifolia</i> *, and <i>Oscillatoria</i>	Alkaloid (phenolic bislactone)	Inflammatory agents, protein kinase C activators. Dilatation of the lymphatic vessel and congestion of capillaries, diarrhoea, and Fibrin deposition in the dilated pulmonary artery followed by bleeding	Fujiki et al. 1983; Wiegand And Pflugmacher 2005
Endotoxins	All cyanobacteria	Lipopolysaccharide	Inflammatory agent, gastrointestinal irritants	Stewart et al. 2006
Lipopolysaccharide		-	-	Metcalfe and Codd 2012
Hepatotoxins	<i>Planktothrix agardhii</i> and <i>P. rubescens</i>	-	-	Dittmann And Wiegand 2006
Microcystins	<i>Anabaena</i> , <i>Anabaenopsis</i> , <i>Hapalosiphon</i> , <i>Microcystis</i> , <i>Nostoc</i> , <i>Oscillatoria</i> , and <i>Planktothrix</i>	Cyclic heptapeptide	Inhibition of protein phosphatases (PP1 and PP2A)	Bagu et al. 1997
Nodularin	<i>Nodularia</i>	Cyclic pentapeptide	Inhibition of protein phosphatases (PP1 and PP2A)	Namikoshi et al. 2003
Neurotoxins	<i>Anabaena</i> , <i>Aphanizomenon</i> , <i>Cylindrospermum</i> , <i>Oscillatoria</i> , <i>Phormidium</i> , and <i>Raphidiopsis</i>	Alkaloid	Binds irreversibly to the nicotinic acetylcholine receptors	Matsunaga et al. 1989
Anatoxin-a (s)	<i>Anabaena</i> , <i>Microcystis</i> , <i>Nostoc</i> , and <i>Planktothrix</i>	Guanidine methyl	Inhibits acetylcholinesterase activity	Lobner et al. 2007
β -N-methylamino-L-alanine (BMAA)	<i>Microcystis</i> and <i>Planktothrix</i>	Phosphate ester	Causes disorder of Motor system, glutamate agonist, increasing the intracellular concentration of calcium in neurons and inducing neuronal activity by hyperexcitation	Faltermann et al. 2014
Cyanopeptolin	<i>Anabaena flos-aquae</i> , and <i>Anabaena lemmermanni</i>	Carbamate alkaloid	Transcriptional alterations of genes belonging to DNA damage and repair	Strichartz et al. 1986
Saxitoxins	<i>Anabaena</i> , <i>Aphanizomenon</i> , <i>Cylindrospermopsis raciborskii</i> , <i>Lyngbya</i> , and <i>Planktothrix</i>	Polyacetates	Binds and blocks the sodium channels in neural cells	Fujiki et al. 1983
Other toxins	<i>Lyngbya majuscula</i>	Lipopeptide	Tumor promoters	Edwards et al. 2004
Debromoaplysitoxin				
Kalkitoxin	<i>Lyngbya majuscula</i>		Block the sodium channels of nerve cell	

Organisms that are eukaryotic algae are represent with “*” mark

Cyanobacterial metabolites and its antiviral potential

Cyanobacterial metabolites work against viruses *via* a variety of different molecular pathways. Inhibiting viral entry by attaching to glycoproteins on viral envelopes or host cell receptors, several cyanobacterial lectins and sulfated polysaccharides stop virus–cell adhesion and membrane fusion. Viral entrance into host cells is prevented, for instance, by cyanovirin-N strong affinity for high-mannose oligosaccharides on viral envelope proteins. Other cyanobacterial metabolites affect viral protein synthesis, destabilize viral capsid structures, or limit reverse transcriptase function, all of which hinder viral replication. When compared to traditional small-molecule antivirals, these mechanisms make cyanobacterial metabolites especially appealing as antiviral options since they target the early stages of infection and may lessen the chance of resistance development. For the past two decades, the search for novel antiviral drug candidates has increasingly focused on cyanobacteria, particularly for combating various viral infections, including coronaviruses (Shalaby And Dubey 2018; Zainuddin et al. 2007). Cyanobacteria and algae are known to produce lectins, naturally occurring proteins with significant antiviral properties. These lectins have demonstrated strong antiviral activity against viruses like Ebola, SARS, and HIV. Study reported by Huskens and Schols (2012), HIV and HSV were used as test models to evaluate antiviral efficacy, and lectins showed inhibitory capacity against both viruses. Moreover, several cyanobacterial metabolites such as cyanovirin-N, phycobiliprotein, calcium spirulan, allophycocyanin, nostoflan, dolastatin-3, ichthyopeptins A/B, microvirin, and lipoproteins exhibit a broad spectrum of antiviral potential. These compounds act by preventing viral entry into epithelial cells of the urogenital tract or by demonstrating virustatic or virucidal activity. Table 4 provides a detailed overview of these metabolites, their respective cyanobacterial sources, and their specific antiviral effects.

The antiviral properties of cyanobacterial polysaccharides

Cyanobacteria have emerged as a pivotal source for antiviral agents, with their active components showing significant potential in combating viral infections. Reichert et al (2017) investigated the antiviral properties of *Arthrospira platensis* polysaccharides, using the supernatant to test its effects on Koi Herpes Virus (KHV), further validating the role of cyanobacteria in antiviral research. Influenza viruses, which cause seasonal epidemics primarily affecting the respiratory system, are responsible for severe health complications

such as pneumonia, pulmonary disease, and cardiovascular issues. Despite the global spread of these viruses, only a limited number of cyanobacterial exopolysaccharides have shown promising effect in prevention and treatment of influenza and pneumonia viral infections.

Nostoflan, an acidic polysaccharide extracted from *Nostoc flagelliforme*, shows notable antiviral properties, particularly against viruses that utilize carbohydrates as cellular receptors (Whitton And Potts 2007). A study conducted in 1997 highlighted the antiviral efficacy of calcium spirulina, derived from hot water extracts of *Spirulina platensis*, against herpes simplex virus (HSV), human immunodeficiency virus type 1 (HIV-1), and human syncytium-forming virus (HSP1). Even at lower dosages, calcium spirulina effectively inhibits viral entry into host cells and the formation of syncytia (Hernández-Corona et al. 2002). Furthermore, its antiviral activity extends to enveloped viruses, with significant effects observed against influenza A virus (IAV) (Ahmadi et al. 2015; Kanekiyo et al. 2007). Similarly, the cyanobacterium, *Aphanothece halophytica* is rich in sulfated exopolysaccharides comprising a diverse array of sugars, including arabinose, rhamnose, fucose, mannose, glucose, galactose, glucuronic acid, and sulfuric acid (Kumari et al. 2020). These compounds have shown promising effects as therapeutic agents in murine models of pneumonia, exhibiting not only antiviral properties but also the ability to mitigate immune system degeneration in affected mice.

Cyanobacterial polysaccharides mainly work with host cell receptors and viral envelope proteins to produce antiviral and immunomodulatory effects (Wu et al. 2020). By imitating host cell surface glycosaminoglycans, sulfated polysaccharides can prevent viral attachment and penetration and obstruct virus–host interactions (Kanekiyo et al. 2007). These polymers may improve innate immune responses and lessen inflammation in infected tissues. Cyanobacterial polysaccharides are attractive prospects for antiviral and nanobiotechnological applications due to their high biocompatibility and structural diversity (da Silva And Teixeira 2024).

Cyanobacterial proteins as antiviral agents

Proteins, like polysaccharides, have the potential to inhibit viral replication. Among these, lectins have been extensively identified and isolated from various cyanobacterial species. Lectins are carbohydrate-binding proteins that interact with glycoproteins or glycol-conjugated carbohydrates (Huskens et al. 2010). Their unique ability to bind directly to the carbohydrates or carbohydrate moieties enables them to play a critical role in antiviral defence mechanisms. When applied

Table 4 Diverse biological functions of cyanobacterial metabolites as antiviral agents (adopted from Prabhu et al. 2022)

Metabolites	Species	Viruses	Mode of study	Mode of action	References
Lectins	<i>Nostoc ellipsoforum</i>	HSV-2, HVP	<i>In vitro</i> ; TZM-bl	Prevent the entrance of viruses into the body via urogenital epithelia	Peeri et al. 2020; Mimount et al., 2012; Li et. al., 2008
Cyanovirin-N	<i>Nostoc ellipsoforum</i>	HIV	<i>In vitro</i> (PBL, MAC)	Virustatic	Levendosky et al. 2015
Phycobiliprotein	<i>Spirulina platensis</i>	MS-2 ΦX-174	<i>In vitro</i> (<i>E. coli</i>)	Virucidal	Boyd et al. 1997
Calcium spirulan	<i>Arthrospira platensis</i>	HSV-1, CTMV, MeV, MuV, IAV, HIV-1	<i>In vitro</i> (Vero, MDCK, HeLa, HEL, MT-4) and <i>In vitro</i> (Vero)	Prevent virus adhesion to host cells	Abd El Hamid et al. 2019; Hayashi et al. 1996
Allophycocyanin	<i>Arthrospira platensis</i>	Enterovirus-71 (EV71)	<i>In vitro</i> (Vero, humanrhabdomyosarcoma)	Prevent virus adhesion to host cells	Lee et al. 2001
<i>Arthrospira</i> crude extracts	<i>Arthrospira platensis</i>	HSV-1	<i>In vitro</i> (Vero)	Prevent virus adhesion to host cells	Shih et al. 2003
Nostoflan	<i>Nostoc flagelliforme</i>	HSV-1	<i>In vitro</i> (Vero)	Prevent the virus adhesion to host cells	Sharaf et al. 2010
Dolastatin 3	<i>Lyngbya majuscula</i>	HIV-1	Enzyme assay	Anti-HIV-1 integrase (inhibit the viral replication cycle)	Kanekiyo et al. 2007
Cyanobacterial extracts	<i>Nostoc, Phormidium, Oscillatoria, Chroococcus, Schizothrix, Aphanocapsa, Synechococcus, Aphanothece, Xenococcus</i>	AMVHIV-1	Enzyme assay	Anti-RT (inhibit the viral replication cycle)	Buffa et al. 2009
Ichthyopeptins A/B	<i>Microcystis ichthyoblabe</i>	IAV	<i>In vitro</i> (MDCK)	Protease inhibition (inhibit viral processing after replication)	Reichert et al. 2017
Cyanovirin-N	<i>Nostoc ellipsoforum</i>	HIV-1	<i>In vitro</i> (CD4+ HeLa)	Anti-synecytia formation (Inhibit the viral release)	Levendosky et al. 2015
Cyanovirin-N	Not reported	HIV-1	<i>In vitro</i> (TZM-bl)	Anti-synecytia formation (Inhibit the viral release)	Zainuddin et al. 2007
Scytovirin	<i>Scytonema varium</i>	EVD	<i>In vivo</i>	Inhibit the virus penetration to the host cell	Garrison et al. 2014
Microvirin	<i>Microcystis aeruginosa</i>	HIV-1	<i>In vitro</i> (CD4+ T-cells)	Anti-synecytia formation (Inhibit the viral release)	Huskens And Schols 2012
Lipoprotein	<i>Arthrospira platensis</i>	IAV	<i>In vivo</i> (murine)	Decreased histopathology	Pugh et al. 2015
Lipoprotein	<i>Arthrospira platensis</i>	(H1N1)	<i>In vitro</i> (THP-1)	Upregulate IL-1β, TNF-α, (IL)-8, MCP-1, MIP-1, MIP-1, IP-10, COX-2	Pugh et al. 2001; Grzanna et al. 2006

EVD Ebola virus disease, HIV Human immunodeficiency virus, HSV Herpes simplex viruses, HVP Human papillomavirus, IAV Influenza A virus, TNF-α Tumor necrosis factor-α, IP-10 Interferon gamma-induced protein 10, MCP-1 monocyte, hemo attractant protein-1, MIP-1 macrophage inflammatory protein-1, IL-18 Interleukin-18, and IL-8 Interleukin 8

topically as microbicides, lectins have demonstrated significant potential in suppressing viral replication, making them promising candidates for the development of novel antiviral therapies.

Cyanovirin-N

Cyanovirin-N, a glycoprotein derived from HIV gp120, consists of 101 amino acids and has a molecular weight of 11 kDa. *In vitro* studies have demonstrated that cyanovirin-N effectively suppresses several HIV laboratory strains and primary isolates, exhibiting LC_{50} values below 5 nmol/L. Its antiviral mechanism is primarily driven by an irreversible binding to HIV gp120, which prevents the virus from attaching to target cells. The cyanovirin-N and gp120 complex has proven to be resistant to denaturing agents, giving cyanovirin-N a strong virustatic effect against HIV virions (Boyd et al. 1997). Another study reported that cyanovirin-N from *Nostoc ellipsosporum* significantly reduced HIV infection in *Rhesus macaques* by 63–85%. *In vitro*, cyanovirin-N consistently inactivated HIV virions with minimal cytotoxicity, highlighting its potential as a promising antiviral agent for HIV treatment (Huskens et al. 2010).

Scytovirin

Scytovirin, a 95-amino acid protein, has demonstrated inhibitory effects against a range of viruses, including HIV, the Zaire ebolavirus, Marburg virus, and SARS-CoV (Li et al. 2008; Bokesch et al. 2003). In studies involving Ebola-infected mice, scytovirin administered at a dose of 30 mg/kg every 6 h resulted in the survival of 9 out of 10 mice, highlighting its potent antiviral potential. Isolated from *Scytonema varium* using an aqueous extraction method, scytovirin was found to contain agglutinin, a protein that binds strongly to glycoproteins on viral envelopes, thereby blocking viral entry into target cells (Sato et al. 2007). Additionally, *Oscillatoria agardhii*, another cyanobacterial species, has also been associated with antiviral compounds. A patent by O'Keefe et al (2015) revealed that scytovirin holds promising effect as a novel antiviral agent, particularly in the treatment of infections caused by high-mannose-enveloped viruses, such as hepatitis C virus (HCV).

Phycobiliproteins

Water-soluble pigment–protein complexes known as phycobiliproteins serve as accessory light-harvesting elements in cyanobacterial photosynthetic machinery. These proteins, which are arranged on the surface of the thylakoid membrane within phycobilisomes, effectively absorb light energy and transmit it to chlorophyll *a* during photosynthesis. The

antioxidant, anti-inflammatory, and antiviral qualities of phycobiliproteins have garnered attention in addition to their physiological function in light harvesting, making them attractive options for use in biotechnology, nutraceuticals, and biomedical research (Kumar et al. 2022; Mogany et al. 2018; Saad et al. 2022). *In vitro* study investigated the antiviral activity of *A. platensis* extract against the bacteriophages MS2 and phiX174. MS2, a single-stranded RNA virus, is commonly used as a model for human poliovirus, hepatitis A virus, and enteroviruses, while phiX174, a single-stranded DNA virus, simulates hepatitis B, HCV, and HIV infections (Bello-Morales et al. 2022). The study revealed that phycobiliprotein-containing extracts from *A. platensis* exhibited a strong virucidal effect on both phages by altering the structure of their viral capsids, thus preventing their integration into *E. coli*. Notably, both phages in this study lack lipid envelopes, similar to several other viruses examined in prior research, suggesting that the virucidal action results from the direct interaction of phycobiliproteins with the protein capsid.

Discovering new anti-cancer compounds through microalgal uniqueness

Since cyanobacterial metabolites can interact with basic biological processes involved in tumor growth and survival, they have garnered a lot of interest in cancer research (Gonçalves et al. 2024; Joshi And Bharadvaja 2023; Pathak et al. 2021). Numerous cyanobacterial substances have anticancer properties through apoptosis induction, protein synthesis inhibition, microtubule dynamics disruption, or interference with cell-cycle regulation (Palko-Labuz et al. 2020). For example, substances produced from dolastatin prevent tubulin polymerization, which causes mitotic arrest and apoptosis in cancer cells that divide quickly, while cryptophycins have a strong cytotoxic effect by disrupting microtubule assembly. Apratoxins are among the other metabolites that impede the growth of tumor cells by preventing co-translational translocation in the secretory route (Kazemi et al. 2021). By targeting pathways that are challenging to modify with traditional synthetic medications, cyanobacteria are a valuable source of structurally unique compounds.

Quinone-based natural compounds, known as calothrixins, have been isolated from *Calothrix* and exhibit potent antiproliferative property against various cancer cell lines (Mahedi 2022; Robles-Bañuelos et al. 2022; Tan And Phyto 2020). Notably, calothrixin B demonstrated significant efficacy against HCT-116 colon cancer cell line, with an IC_{50} value of 0.32 μ M, indicating its strong antiproliferative potential (Hatae et al. 2014). Other microalgae-derived compounds have also shown promising effect in targeting

colon cancer. For instance, marine cyanobacteria produce malyngamides, small amides that include malyngamide C and 8-epi-malyngamide C, which exhibited cytotoxic effects against HT-29 colon cancer cells with IC_{50} values of 5.2 and 15.4 μ M, respectively. Mero cyclophanes A and B, isolated from a terrestrial *Nostoc* sp. (UIC 10022A), also demonstrated significant antiproliferative property against HT-29 cells, with an IC_{50} values of 3.3 and 1.7 μ M, respectively (Kang et al. 2012). Furthermore, the secondary metabolite hierridin B, a polyketide isolated from *Cyanobium* sp. LEGE 06113, exhibited selective cytotoxicity against HT-29 cells, with an IC_{50} value of 0.1 mM (Leao et al. 2013).

Glembatumumab vedotin, a cyanobacterial-derived peptide currently in Phase II clinical trials, has shown significant efficacy in the treatment of breast cancer and melanoma. Another promising class of cyanobacterial compounds are apratoxins, cyclic depsipeptides first discovered in *Micronesia* (Luesch et al. 2006). Apratoxin A, a key member of this group, has been shown to induce apoptosis and induce a distinct G1 cell cycle arrest. Moreover, apratoxin A effectively inhibits secretory pathway through co-translational translocation blockage. Several bilirubin-like compounds derived from *S. platensis* have been reported to exhibit antitumor activity, particularly against pancreatic cancer (Konícková et al. 2014). Similarly, largazole, a compound isolated from *Symploca* sp., has demonstrated significant antiproliferative effects due to its unique chemical structure (Taori et al. 2008). Apratoxins shows their potent cytotoxicity against various cancer cell lines at nanomolar concentrations. For example, apratoxin A, derived from the cyanobacterial strain *Lyngbya bouillonii*, exhibits cytotoxic activity against adenocarcinoma cells. Additionally, coibamide, isolated from *Leptolyngbya* sp., has demonstrated toxicity against mouse neuro-2a and NCI-H460 lung cancer cells (Medina et al. 2008).

Numerous cyanobacterial compounds exhibit potent cytotoxicity, a characteristic that has garnered significant attention in cancer research (Gerwick et al. 2001). Notably, borophycin, cryptophycins 1 and 8, and cyanovirin are among the most significant discoveries in this context. Borophycin, a boron-containing metabolite, was isolated from *Nostoc linckia* and *Nostoc spongiforme* (Banker And Carmeli 1998). This compound has demonstrated cytotoxic effects against human epidermoid carcinoma (LoVo) and human colorectal adenocarcinoma (KB) cell lines. Violaxanthin identified as pro-apoptotic and antiproliferative activities against various human cancer cell lines (Pasquet et al. 2011). Notably, the natural oxadiazine nocuolin A (NoA), characterized by its unique 1,2,3-oxadiazine structure, has been isolated from several cyanobacterial genera, including *Nostoc*, *Nodularia*, and *Anabaena*. These genera have demonstrated biodiversity of natural products, as evidenced by

the identification of putative gene clusters associated with NoA biosynthesis in their genomes (Voráčová et al. 2017). Research indicates that NoA-induced cell death exhibits characteristics consistent with caspase-dependent apoptosis. Interestingly, NoA demonstrates anti-proliferative effects against a range of human cancer cell lines, particularly those harboring mutations in the p53 tumor suppressor gene, with IC_{50} values ranging from 0.7 to 4.5 μ M (Voráčová et al. 2017). NoA as a member of a novel class of heterocyclic natural metabolites, highlighting its potential as a therapeutic agent in cancer treatment.

Historically, similar heterocyclic structures have primarily been confined to synthetic compounds, particularly those based on 1,3,4- or 1,2,4-oxadiazines, where a carbon atom is strategically incorporated between two nitrogen atoms or between a N and an O atom, effectively disrupting the N–N–O linkage system (Bajaj et al. 2015). The only documented cases of actual N–N–O linkages occur within 5 membered rings, specifically in 1,2,3-oxadiazole derivatives (Semenov And Makarova 2011). Initially, the existence of a closed 1,2,3-oxadiazole ring was deemed improbable, as quantum chemical modelling predicted that an open-ring form would be more stable (Nguyen et al. 1985). However, incorporation of specific substituents can shift the equilibrium towards the cyclic form, as evidenced by various synthetic 1,2,3-oxadiazole derivatives (Todres 2011). The discovery of nocuolin A (NoA) offers novel insights into the chemistry of heterocyclic structures featuring N–N–O linkages, revealing their occurrence in natural products. This is particularly significant from a pharmaceutical perspective, as 1,3,4-oxadiazoles represent a substantial platform for drug development (Bajaj et al. 2015). The interest in synthesizing new 1,3,4-oxadiazole derivatives is growing, driven by their demonstrated wide-ranging biological activities. Notably, synthetic 1,3,4-oxadiazine derivatives have exhibited high efficacy against cancer cells in the nano-molar range (Mohareb And Schatz 2011).

Techniques for solving problems: creation of high potency drug delivery formula possibility of microalgal drug development using peptides

Peptide conjugation strategies have been leveraged in the pharmaceutical industry to address the challenges of stability and solubility associated with natural compound-based drug development (Chuang et al. 2009; Anand et al. 2023; Ibeanu et al. 2020; Karki et al. 2020). Extensive research has been focused on identifying novel treatments for malignant cells, utilizing natural peptides from diverse sources as well as their synthetic analogs (Thundimadathil 2012; Singh

et al. 2011; Tabassum and Ahmad 2022). Cationic antitumor peptides, in particular, have garnered attention as promising agents for anticancer therapies, owing to their numerous advantages over traditional chemical agents (Alas et al. 2020). These include a low molecular mass, relatively straightforward structures, heightened specificity in targeting tumor cells compared to healthy cells, a reduced propensity to induce multidrug resistance, and favorable absorption characteristics (Liu et al. 2013; Mader And Hoskin 2006). When used in combination with conventional therapeutic agents, these peptides have the potential to enhance drug efficacy, thereby improving treatment outcomes (Chuang et al. 2009).

Currently, there are approximately 50 to 60 approved peptide drugs on commercial platforms, with expectations for this number to increase in the coming years. Among these, three of the four peptide drugs utilized for cancer treatment leuprolide acetate (Lupron), goserelin acetate (Zoladex), and octreotide acetate (Sandostatin) have collectively achieved global sales exceeding \$1 billion USD (Reichert et al. 2010; Thundimadathil 2012). Notably, around 30 peptides have been discovered as peptide-based anticancer drugs, establishing a foundational platform for cancer therapy (Mendoza et al. 2005). Combining bioactive peptides can improve stability against enzymatic degradation, boost formulation flexibility, and create synergistic interactions that promote medicinal effectiveness (Wang et al. 2021). Multiple molecular pathways may be targeted simultaneously by synergistic peptide combinations, which could lower the dosage requirement and prevent the development of resistance. Furthermore, in pharmaceutical formulations, peptide combinations can enhance pharmacokinetic characteristics, solubility, and delivery effectiveness. This approach may offer advantages over single isolated microalgal compounds. However, peptide-based developments are not without their drawbacks, including poor bioavailability, instability, insolubility in natural compounds, and potential toxicity to host cells (Reichert et al. 2010). These challenges could significantly hinder the application of cost-effective production methods for marine-derived compounds in commercial platforms.

Nano formulation based on microalgal polysaccharides

Cyanobacterial polysaccharides are becoming more and more popular as biocompatible materials for use in nanobio-technology. Essential characteristics of these extracellular and intracellular polysaccharides include their low toxicity, biodegradability, structural flexibility, and capacity to produce hydrogels and nanoparticulate systems. Cyanobacterial

polysaccharides are therefore attractive options for environmental applications, medication administration, and the production of biomaterials.

Cyanobacterial polymer-based nanopolysaccharide systems can serve as carriers for bioactive substances, enhancing their solubility, stability, and controlled release. Their functional groups facilitate cross-linking and chemical manipulation, enabling the creation of nanoparticles with specific physicochemical characteristics. Polysaccharide-based nanoparticles have been investigated in biomedical settings for antibacterial formulations, antioxidant delivery, and medicinal chemical encapsulation. Numerous studies have investigated polysaccharide-based nanomaterials for a variety of biomedical applications, including antimicrobial activity, drug and gene delivery, tissue engineering, cancer therapy, and wound healing. Notably, research indicates that cyanobacteria secrete polysaccharides, which exhibit diverse biological effects, including antiviral activity (Kanekiyo et al. 2005). Cyanobacterial polysaccharides released by *A. platensis* have demonstrated *in vitro* and *in vivo* antiviral activity against two strains of vaccinia virus and *Ectromelia* virus, highlighting their potential as a source of biologically active compounds for therapeutic applications.

A novel acidic polysaccharide known as nostoflan, derived from *N. flagelliforme*, has exhibited significant antiviral potential against various enveloped viruses, including the influenza virus (Kanekiyo et al. 2005). Furthermore, compounds that scavenge free radicals, such as sulfated polysaccharides isolated from cyanobacteria, have been shown to mitigate cancer development. Additionally, polysaccharides extracted from *Spirulina* have demonstrated a range of biological activities, with research indicating that sulfated polysaccharides inhibit tumor cell proliferation both *in vitro* (Jia et al. 2008) and *in vivo* (Akao et al. 2009). Since polysaccharide-lipid hybrid nanoparticles are commonly employed in drug delivery systems, lipid components are pertinent even though this section concentrates on nanopolysaccharides. Several microalgal species, including *Chaetoceros*, *Isochrysis*, *Nannochloropsis*, *Pavlova*, *Phaeodactylum*, *Skeletonema*, *Thalassiosira*, and *Tetraselmis*, have shown promising effect in animal nutrition due to their rich content of biologically active essential lipids, such as omega-3 and omega-6 fatty acids (Muller-Feuga 2000). Despite the compositional differences among microalgae, their bioactive components may yield enhanced benefits when integrated with other nanostructures, suggesting a valuable avenue for advancing nutraceutical and therapeutic applications.

Cyanobacterial metabolite research: computational and analytical tools

Identification and measurement of metabolites are key components of metabolite analysis, which is an essential part of comprehending biological systems. Numerous tools and software have been created to aid in this process, allowing researchers to quickly evaluate and comprehend intricate metabolomics data.

Cyanobacterial metabolite and biosynthesis gene cluster databases

Publicly accessible databases, which offer curated genomic, biochemical, and metabolite data, are essential to the study of cyanobacterial metabolites. These databases, including GenBank, KEGG, UniProt, and MIBiG, are fundamental tools for the functional characterisation and computational identification of cyanobacterial metabolites.

In order to properly convey complicated data insights, data visualization is a crucial component of metabolomics research. To make data visualization easier, a number of software applications have been created, including ggplot2, a well-liked R data visualization program with features including editable graphs and plots, Matplotlib is a feature-rich Python data visualization package that offers resources for producing excellent 2D and 3D charts. Cytoscape is a network visualization software platform that helps scientists see and examine intricate biological networks (Shannon et al. 2003). With the aid of the aforementioned software and tools, researchers can identify metabolites i.e., find out if biological samples contain metabolites and in what quantities; analyse metabolic pathways means recognize important pathways involved in different biological processes and comprehend metabolic networks and data visualization to make metabolomics data easier to understand and communicate, create charts and networks.

A curated database devoted to cyanobacterial secondary metabolites, CyanoMetDB offers physicochemical characteristics, molecular formulas, chemical structures, and literature references for cyanometabolites that have been described. This website facilitates comparative chemical studies, metabolite identification, and dereplication in cyanobacterial research (Jones et al. 2021). The term “CyanoMet Mass” most likely describes a mass spectrometry-based method for examining cyanobacterial metabolites. In complicated biological systems, mass spectrometry is a potent analytical method for identifying and measuring metabolites. This method can assist researchers in finding new molecules with potential uses in biotechnology, agriculture, or pharmaceuticals by identifying novel

metabolites, describing metabolic pathways means recognizing how metabolites are produced and regulated in cyanobacteria, and calculate the metabolite abundance vary in response to genetic or environmental changes. Hence, by using mass spectrometry to investigate the metabolism of cyanobacteria, researchers can learn more about the biology and chemistry of these organisms, which will ultimately help with applications in industries like biotechnology and medicine.

Structures, taxonomy of producing organisms, classifications of biosynthetic and chemical ontologies, categorization by compound classes, and cross-links to a variety of other natural product-related data resources are all included in the Natural Products Atlas, a database of natural products created from microbes. A comprehensive database and tool, the Natural Product Atlas was created to make it easier to find, annotate, and analyze natural products. Researchers may examine the chemical structures, biological activity, and biosynthetic routes of a large number of known natural compounds. Some important characteristics are Database search: By using criteria like name, structure, or biological activity, users can look for certain natural items. Structure analysis: The platform provides capabilities for predicting attributes and analyzing chemical structures. Analysis of biosynthetic routes: Scientists can investigate the biosynthetic pathways connected to natural products. By the use of the Natural Product Atlas, the researchers are able to find new substances and investigate their possible uses and look into the biological processes and modes of action of natural substances. Through its extensive and intuitive interface, the Natural Product Atlas speeds up natural product chemistry research, enabling advancements in industries including biotechnology, agriculture, and medicines (Poynton et al. 2025). Further, PubChem is an extensive online database of chemical compounds that gives users access to a wealth of data about molecules, such as their characteristics, structures, and biological activities. PubChem facilitates advancements in drug development, environmental science, and public health by accelerating research in chemistry, biology, and allied subjects through the provision of an extensive and freely available database (Kim et al. 2025).

The Norman Network ALGALTOX List tool is a thorough database or platform that offers vital information about algal toxins, supporting environmental monitoring, public health protection, and research. It provides materials on detection techniques for biological and aquatic samples, along with in-depth information on the structures, characteristics, and toxicity profiles of various algal toxins. The ALGALTOX List application centralizes this data, making it accessible to researchers, environmental scientists, and public health experts. This facilitates advancements in aquatic ecosystem management, environmental science,

and public health. The ecosystem and human health can be shielded from the harmful impacts of algae toxins by using this tool to guide mitigation methods and inform risk assessments (US Epa 2019).

The resource known as the NORINE database is basically devoted to nonribosomal peptides (NRPs), a broad category of bioactive substances produced by non-ribosomal peptide synthetases (NRPS). With the help of this platform researchers investigate the structures, characteristics, and biosynthetic mechanisms of NRP. Natural product discovery, pharmaceutical research, and biocatalysis are all made easier by NORINE central NRP research hub, which eventually speeds up biotechnology and treatment development (Caboche et al. 2007; Flissi et al. 2020). Furthermore, a comprehensive collection of natural products, the COCONUT (COllection of Open Natural prodUcTs) database was launched in 2021 which provides researchers with a single platform to access in-depth details about their structures, characteristics, and biological activities (Sorokina et al. 2021). COCONUT supports biotechnology, pharmaceutical research, and natural product discovery using capabilities like database search, structural analysis, and biological activity data. Scientists can discover new chemicals, look into their potential as medicines, and examine biotechnological uses. In COCONUT database, abundance of natural product data propels advancements in agriculture, medicine, and related sectors. This database bridges the gap between the discovery of natural products and real-world applications by speeding up research and innovation (Chandrasekhar et al. 2025).

Analytical and genome-mining tools

Cyanobacterial secondary metabolite identification, annotation, and characterisation are made possible by analytical and genome-mining methods. Biosynthetic gene clusters can be automatically identified from genomic data using platforms like antiSMASH. Redundancy in natural product discovery is decreased using metabolomics technologies that facilitate dereplication, spectrum comparison, and metabolite classification, such as GNPS molecular networking and MetaboAnalyst. These techniques serve as a link between computational annotation and experimental metabolomics (Medema et al. 2011). The mass spectrometry is a popular method for analysing metabolites, a number of software applications have been created to process and examine data from this method. Among them are XCMS, a well-liked software program for handling and evaluating mass spectrometry data, especially for untargeted metabolomics research. Whereas, Mzmine is a mass spectrometry data analysis software platform with capabilities like

identification, alignment, and peak detection (Katajamaa et al. 2006; Pluskal et al. 2010). MassHunter is a mass spectrometry data collection and analysis software program that offers capabilities for compound identification and quantitative analysis.

Another effective method for metabolite analysis is NMR spectroscopy, and a number of software applications have been created to help with the processing of NMR data (Krylova And Kurashov 2022). These consist of Chenomx NMR Suite, is a software program for analysing NMR data that includes functions including component identification, quantification, and spectral processing while, MetaboMiner is a software program for metabolomics based on NMR that helps scientists find and measure compounds in intricate biological materials (Xia et al. 2008).

Interpreting metabolomics data requires an understanding of metabolic networks and pathways. To make metabolic pathway analysis easier, a number of software tools have been created, including MetaboAnalyst, an online tool for analysing metabolomics data that provides network visualization, pathway analysis, and enrichment analysis (Xia And Wishart 2016). Researchers may find and examine metabolic pathways in a variety of organisms by using KEGG, a comprehensive database of metabolic processes. Reactome is a database of biological pathways, including metabolic pathways, that offers visualization and analytical tools (Fabregat et al. 2018).

Artificial intelligence (AI) based methods for cyanometabolite discovery

The discovery of natural products, particularly studies on cyanobacterial secondary metabolites, has been profoundly altered by recent developments in artificial intelligence (AI) and machine learning (Fig. 6). Genome mining is increasingly using AI-driven computational methods, which allow for the quick discovery and annotation of biosynthetic gene clusters (BGCs) that produce metabolites (Blin et al. 2016; Challis 2008; Kenshole et al. 2021; Medema et al. 2011; Weber et al. 2015). In order to predict biosynthetic pathways and related chemical scaffolds from genomic data, platforms like antiSMASH and the Minimum Information about a Biosynthetic Gene cluster (MIBiG) repository combine rule-based algorithms with machine learning techniques. This accelerates in the discovery of novel natural products (Blin et al. 2016; Kautsar et al. 2020; Terlouw et al. 2023).

In metabolomics, AI-based techniques are being used more and more for compound dereplication, structure prediction, and annotation of unknown metabolites, in addition to genome analysis. In order to reduce redundancy and

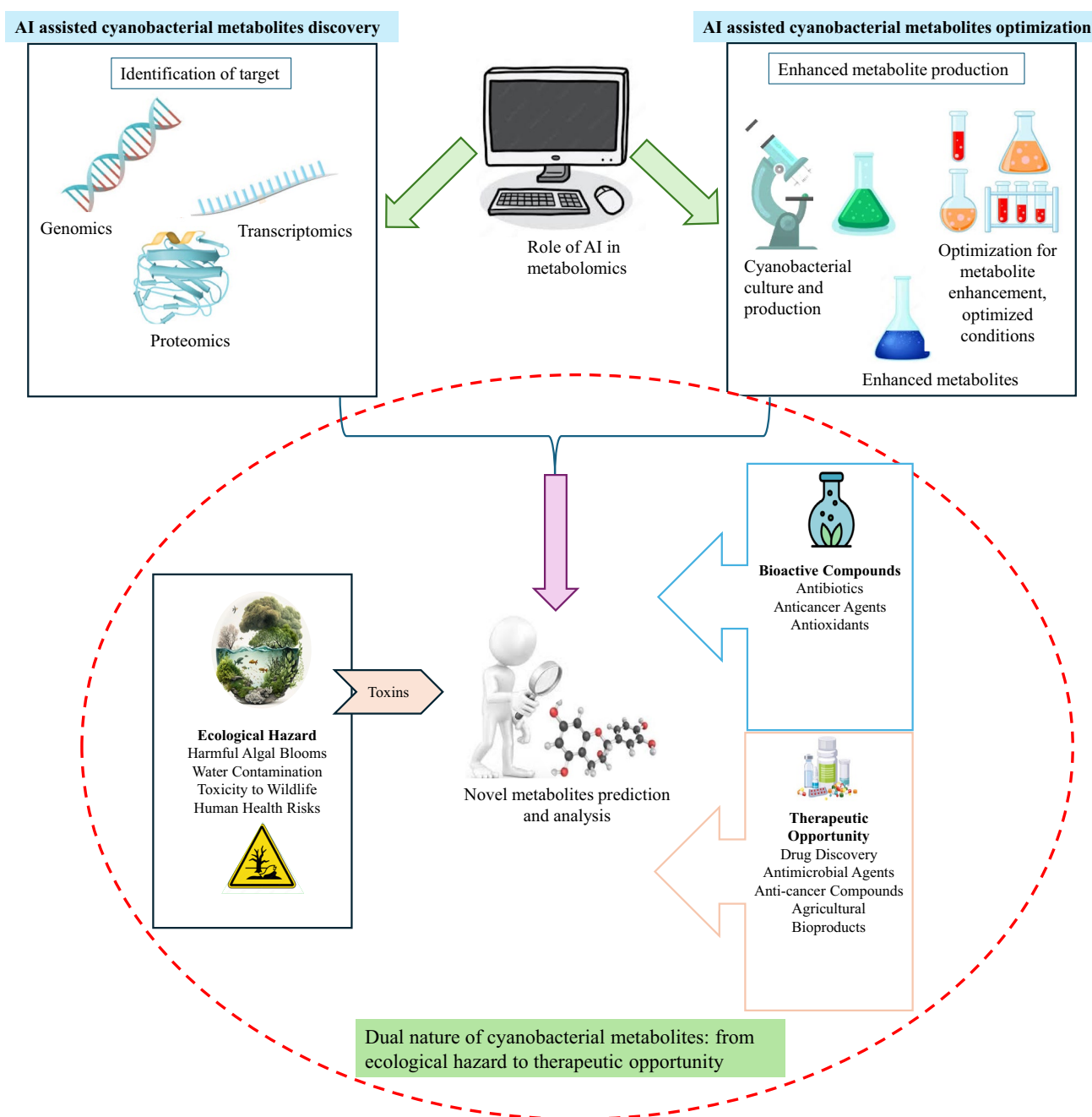


Fig. 6 The schematic diagram represents the dual nature of cyanobacterial metabolites and the AI-assisted cyanometabolite drug discovery

experimental costs, machine learning models trained on large mass spectrometry and nuclear magnetic resonance datasets make it easier to recognize patterns, classify metabolites, and predict structure–activity relationships (Hannigan et al. 2019; Mallowney et al. 2023). Furthermore, deep learning frameworks have been investigated for strain engineering, metabolic flux prediction, and culture condition optimization, all of which enable higher production of bioactive secondary metabolites (Liao et al. 2025; Zhang et al. 2025a, b; Zhang et al. 2026).

Using AI in conjunction with genomics, transcriptomics, and metabolomics in the context of cyanobacteria has great potential to reveal undiscovered biosynthetic pathways, rank candidate metabolites for pharmaceutical screening, and improve sustainable production methods (Lu et al. 2025; Khan And Khan. 2025). In contrast to actinomycetes and fungi, the application of AI to cyanobacterial systems is still in its infancy; however, these computational methods offer a potent complement to biotechnological exploitation

and secondary metabolite discovery (Skinnider et al. 2020, 2021).

Conclusions and future prospects

Cyanobacteria are a metabolically diverse genus of photosynthetic microorganisms that can produce a variety of secondary metabolites with medicinal and ecological significance. These substances come from specific biosynthetic pathways and are often involved in signaling, chemical defense, stress tolerance, and nutrition uptake. Research on cyanobacterial natural products continues to advance by gaining greater understanding of the connections between environmental factors, biosynthetic gene clusters, and metabolite activity. Despite significant advancements, there are still significant gaps in our understanding of cyanobacterial metabolites. The majority of described chemicals come from a small number of taxa, and little is known about the metabolic potential of harsh conditions and understudied cyanobacterial lineages. Furthermore, the molecular targets of many metabolites are still unknown, and many biosynthetic pathways forecasted by genome mining have not been experimentally validated. Future studies should focus on integrative methods that combine ecological analysis, biochemical validation, metabolomics, and genetics to elucidate metabolite function and biosynthetic regulation. It is anticipated that developments in genome mining, computational analysis, and machine learning-based prediction tools would speed up the discovery of metabolites and enhance sustainable production methods. Conclusively, the conversion of cyanobacterial chemical variety into trustworthy biotechnological and medicinal applications will require the integration of ecological context, biosynthetic pathways, and contemporary analytical techniques. In order to facilitate further advancements in the study of cyanobacterial metabolites, this review offers a summarized framework.

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